

Εργαστήριο Σχεδιασμού Ολοκληρωμένων Κυκλωμάτων 2

Ομάδα 10

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Modulo 5 Adder

Σκοπός είναι η υλοποίηση ενός παραμετρικού adder ο οποίος υπολογίζει το υπόλοιπο όταν το άθροισμα δυο N-bit αριθμών διαιρείται με το 5.

Αθροιστές

```

library ieee;
use ieee.std_logic_1164.all;

entity full_adder is
    port(
        a,b,cin: in  std_logic;
        sum,cout: out std_logic
    );
end full_adder;

architecture dataflow of full_adder is
    begin
        sum  <= a xor b xor cin;
    
```

```

    cout <= (a and b) or (a and cin) or (b and cin);
end dataflow;

library ieee;use ieee.std_logic_1164.all;

entity adder_n is generic (n:integer:=8); port(
    a,b : in std_logic_vector (n-1 downto 0);
    cin: in std_logic;
    sum: out std_logic_vector (n-1 downto 0);
    cout: out std_logic); end adder_n;

architecture dataflow of adder_n is

    component full_adder port(
        a,b,cin: in std_logic;
        sum,cout: out std_logic); end component;

    signal wire_carry:std_logic_vector(n downto 0);

begin

    wire_carry(0) <= cin;

    generate_label:
    for i in 0 to n-1 generate
        full_adder_i : full_adder port map
(a(i),b(i),wire_carry(i),sum(i),wire_carry(i+1));
    end generate;
    cout <= wire_carry(n);

end dataflow;

```

Flip Flop

```

library ieee; use ieee.std_logic_1164.all; use ieee.numeric_std.all;

entity qd is generic (n:integer:=8); port (q:in std_logic_vector(n-1
downto 0); clk,rst:std_logic;
                                d:out std_logic_vector(n-1 downto 0));
end qd;

architecture my_arch of qd is begin process(clk,rst)
begin

```

```

    if rst='1' then
        d<=(others=>'0');
    elsif clk'event and clk='1' then
        d<=q;
    end if;
end process;
end my_arch;

```

Συγκριτής

```

library ieee; use ieee.std_logic_1164.all; use ieee.numeric_std.all;

entity comparator_of_5 is generic (n:integer:=4); port(
    S: in std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
);
end comparator_of_5;

architecture dataflow of comparator_of_5 is

    signal above_below : std_logic_vector(1 downto 0);
begin
    above_below <= "10" when (S(n-1 downto 3) /=
std_logic_vector(to_unsigned(0, n-3)) and negative = '0') or
(negative = '0' and S(2) = '1' and (S(1) = '1'
or S(0) = '1')) else
        "01" when (S(n-1) = '1' and negative = '1') else
        "00";
    above <= above_below(1);
    below <= above_below(0);
end dataflow;

```

/comparator_of_5/n	4	4						
/comparator_of_5/S	0011	1000					0011	
/comparator_of_5/negat...	0							
/comparator_of_5/above	0							
/comparator_of_5/below	0							
/comparator_of_5/above...	00	01		10		00		

Υπολογισμός κρατουμένου

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity remainder_calculator is generic (n:integer:=6);
  port (
    data_in      : in  std_logic_vector(n-1 downto 0);
    negative_first: in  std_logic;
    remainder     : out std_logic_vector(n-1 downto 0)
  );
end entity;

architecture rtl of remainder_calculator is
  signal above1, below1 : std_logic;
  signal above2, below2 : std_logic;
  signal temp1, temp2    : std_logic_vector(n-1 downto 0);
  signal stage2_in       : std_logic_vector(n-1 downto 0);
  signal stage2_temp1, stage2_temp2 : std_logic_vector(n-1 downto 0);
  signal data : std_logic_vector(n-1 downto 0);
  signal mod5, mod5_comp : std_logic_vector(n-1 downto 0);
  constant all_ones : std_logic_vector(n-1 downto 0) := (others =>
'1');

  component comparator_of_5 is generic (n:integer:=4); port(
    S: in  std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
  );
end component;

  component adder_n is
    generic (n: integer := 4);
    port(
      a, b: in std_logic_vector(n-1 downto 0);
      cin: in std_logic;
      sum: out std_logic_vector(n-1 downto 0);
      cout: out std_logic
    );
  end component;

begin
  mod5 <= std_logic_vector(to_signed(5, n));
```

```

mod5_comp <= std_logic_vector(to_signed(-5, n));

data <= (others => '0') when (data_in = all_ones and negative_first =
'0') else
    data_in;
-- First comparison stage
comp1: comparator_of_5
    generic map(n)
    port map(data, negative_first, above1, below1);

-- First arithmetic stage
add_above: adder_n
    generic map(n)
    port map(data, mod5_comp, '0', temp1, open);

add_below: adder_n
    generic map(n)
    port map(data, mod5, '0', temp2, open);

-- Intermediate mux
stage2_in <= temp1 when above1 = '1' else
    temp2 when below1 = '1' else
    data;

-- Second comparison stage
comp2: comparator_of_5
    generic map(n)
    port map(stage2_in, negative_first, above2, below2);

-- Second arithmetic stage
add_above2: adder_n
    generic map(n)
    port map(stage2_in, mod5_comp, '0', stage2_temp1, open);

add_below2: adder_n
    generic map(n)
    port map(stage2_in, mod5, '0', stage2_temp2, open);

-- Final output selection
remainder <= stage2_temp1 when above2 = '1' else
    stage2_temp2 when below2 = '1' else
    (others => '0') when (data_in = all_ones and negative_first =
'0') else--for 15
    stage2_in;

```

```
end architecture;
```

/remainder_calculator/n	6	6					
/remainder_calculator/data_in	3	-32				3	
/remainder_calculator/negative...	1						
/remainder_calculator/remainder	3	22		-22		3	
/remainder_calculator/above1	0						
/remainder_calculator/below1	0						
/remainder_calculator/above2	0						
/remainder_calculator/below2	0						
/remainder_calculator/temp1	-2	27				-2	
/remainder_calculator/temp2	8	-27				8	
/remainder_calculator/stage2_in	3	27		-27		3	
/remainder_calculator/stage2_t...	-2	22		-32		-2	
/remainder_calculator/stage2_t...	8	-32		-22		8	

Κύριος Κώδικας mod5 όπου αθροίζονται με διάφορα στάδια τα διάφορα υπόλοιπα των κομματιών

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity mod5 is
    generic (n: integer := 32);
    port(
        A, B: in std_logic_vector(n-1 downto 0);
        remainder: out std_logic_vector(3 downto 0) -- [4-0] signed
    );
end mod5;

architecture dataflow of mod5 is

    signal S: std_logic_vector(n-1 downto 0);
    signal padded_32 : std_logic_vector(31 downto 0);
    signal Module, Module_Complement : std_logic_vector(3 downto 0);

    -- Correct array type declaration
    type slv4_array is array (natural range <>) of std_logic_vector(3
downto 0);
    signal remainders : slv4_array(0 to 7);
```

```

    type slv6_array is array (natural range <>) of std_logic_vector(5
downto 0);
    -- Adder tree signals
    signal step:slv6_array(0 to 3);

    signal sum_stage1 : slv6_array(0 to 3);
    signal sum_stage2 : slv6_array(0 to 1);
    signal sum_final : std_logic_vector(5 downto 0);
    signal sum_final_temp1,sum_final_temp2:std_logic_vector(3 downto 0);
    signal abovesum, belowsum: std_logic;

    -- decide if i want to calculate as if its a negative or positive
number
    signal negative:std_logic;
    signal s_first:std_logic_vector(3 downto 0);
    --Component declarations
    component adder_n is
        generic (n: integer := 4);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
        );
    end component;

    component comparator_of_5 is
        generic (n: integer := 4);
        port(
            S: in std_logic_vector(n-1 downto 0);
            negative: in std_logic;
            above: out std_logic;
            below: out std_logic
        );
    end component;

    component remainder_calculator is generic(n: integer := 32);
    port (
        data_in      : in  std_logic_vector(n-1 downto 0);
        negative_first: in  std_logic;
        remainder     : out std_logic_vector(n-1 downto 0)
    );
    end component;

```

```

begin
Module <= std_logic_vector(to_signed(5, 4));
Module_Complement <= std_logic_vector(to_signed(-5, 4));

adder : adder_n
    generic map(n)
    port map (A, B, '0', S, open);

padded_32 <= (31 downto n => S(n-1)) & S;

negative <= padded_32(31);

s_first <= padded_32(31 downto 28);

remainder_c : remainder_calculator
generic map(4)
port map(s_first,negative,
remainders(7));

generate_label:
for i in 0 to 6 generate

signal s_in: std_logic_vector(3 downto 0);
begin

s_in <= padded_32(3+i*4 downto i*4);

remainder_c : remainder_calculator
generic map(n=>4)
port map(s_in,'0',
remainders(i));
end generate;

-- Adder tree to sum all remainders
-- Stage 1: Add pairs of remainders

gen_stage1: for i in 0 to 3 generate
    signal a_padded, b_padded : std_logic_vector(5 downto 0);
begin
    a_padded <= "00" & remainders(i*2);
    b_padded <= "00" & remainders(i*2+1);

```

```

    adder_st1: adder_n
      generic map(n => 6)
      port map(
        a => a_padded,
        b => b_padded,
        cin => '0',
        sum => sum_stage1(i),
        cout => open
      );
end generate gen_stage1;

-- Stage 2: Add results from stage 1
gen_stage2: for i in 0 to 1 generate
  adder_st2: adder_n
    generic map(6)
    port map(
      a => sum_stage1(i*2),
      b => sum_stage1(i*2+1),
      cin => '0',
      sum => sum_stage2(i),
      cout => open
    );
end generate gen_stage2;

-- Final adder
adder_final: adder_n
  generic map(6)
  port map(
    a => sum_stage2(0),
    b => sum_stage2(1),
    cin => '0',
    sum => sum_final,
    cout => open
  );

step(0) <= sum_final;

generate_remainder_out_of_sum:
for i in 0 to 2 generate

  remainder_c : remainder_calculator
  generic map(n=>6)
  port map(step(i), '0',
    step(i+1));
end generate;

```

```
end dataflow;
```

[illegible]

```

library ieee; use ieee.std_logic_1164.all;

entity mod_flip_flop is generic (n:integer:=32); port(
clk, rst : in std_logic;
A,B: in std_logic_vector(n-1 downto 0);
remainder: out std_logic_vector(3 downto 0)); end mod_flip_flop;

architecture dataflow of mod_flip_flop is

    component mod5 is
        generic (n:integer:=32);
        port(
            A,B: in std_logic_vector(n-1 downto 0);
            remainder: out std_logic_vector(3 downto 0)
        );
    end component mod5;

    component qd generic(n:integer:=32); port(
        q : in std_logic_vector(n-1 downto 0); clk, rst : in
std_logic;
        d : out std_logic_vector(n-1 downto 0)); end component;

```

```

    signal a_ff, b_ff: std_logic_vector(n-1 downto 0);
    signal remainder_ff, remainder_vec: std_logic_vector(3 downto 0);
begin
    --first flip flop
    a_qd: qd generic map(n) port map(a, clk, rst, a_ff);
    b_qd: qd generic map (n) port map(b, clk, rst, b_ff);

    --carry select adder
    adder: mod5 generic map(n) port map(a_ff, b_ff, remainder_vec);

    --second flip flop
    rem_qd: qd generic map (4) port map(remainder_vec, clk, rst,
remainder_ff);
    remainder<=remainder_ff;

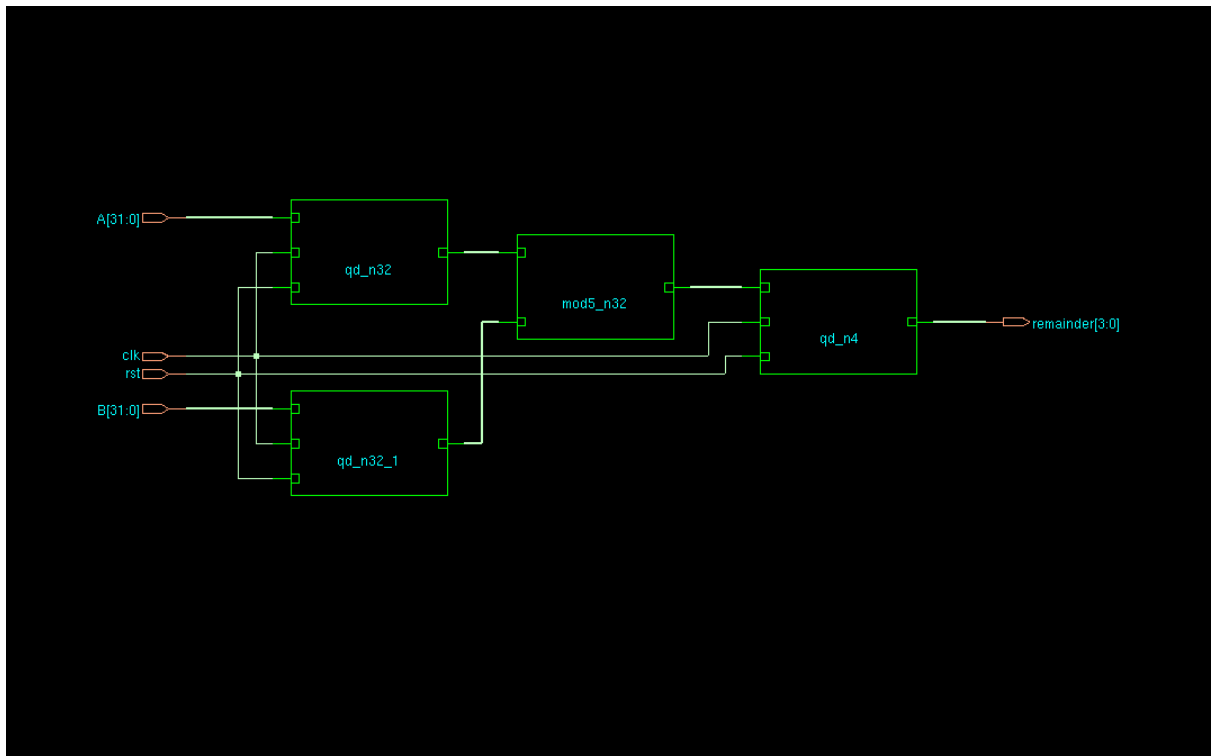
end dataflow;

```

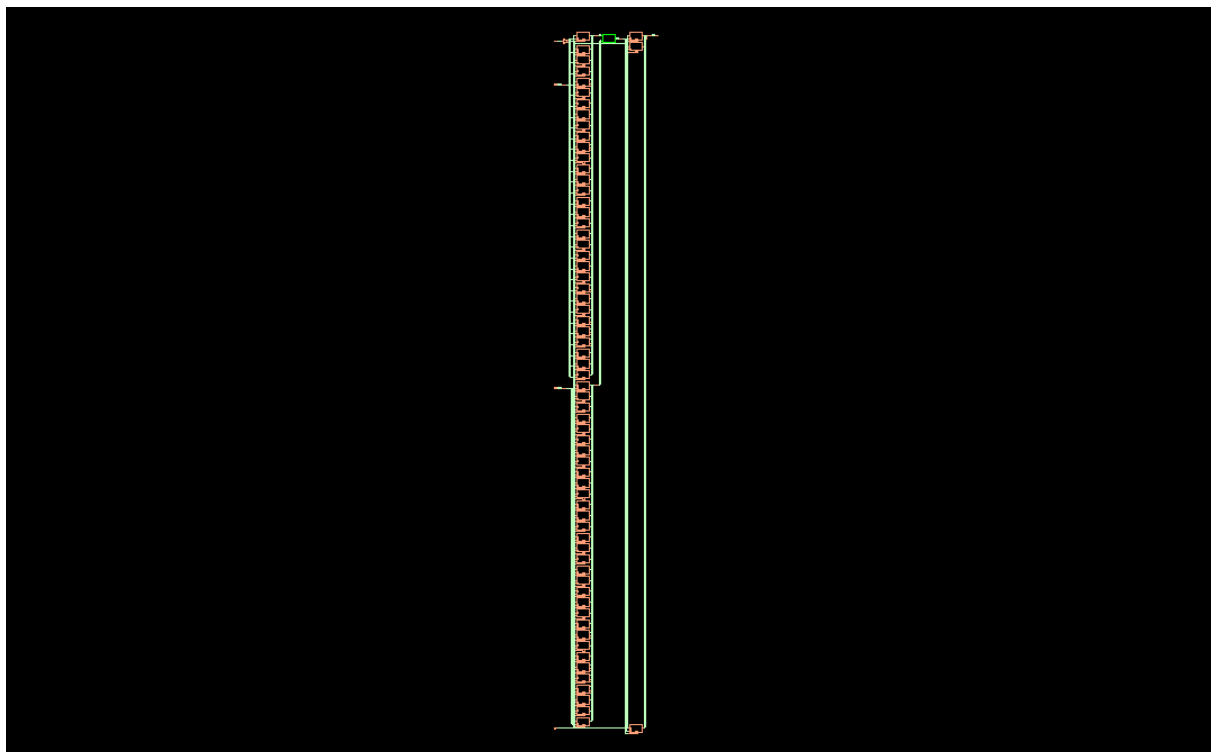
TestBench

❖ /testbenc...	2147483647	216	-92	715827882	-715827883	2147483647
❖ /testbenc...	-1	300	100	357913941	-357913941	-1
❖ /testbenc...	0					
❖ /testbenc...	0					
❖ /testbenc...	1	X	1	3	1	

Genus Modulo 5 Adder με Flip Flops



Genus Modulo 5 Adder με Flip Flops 45nm



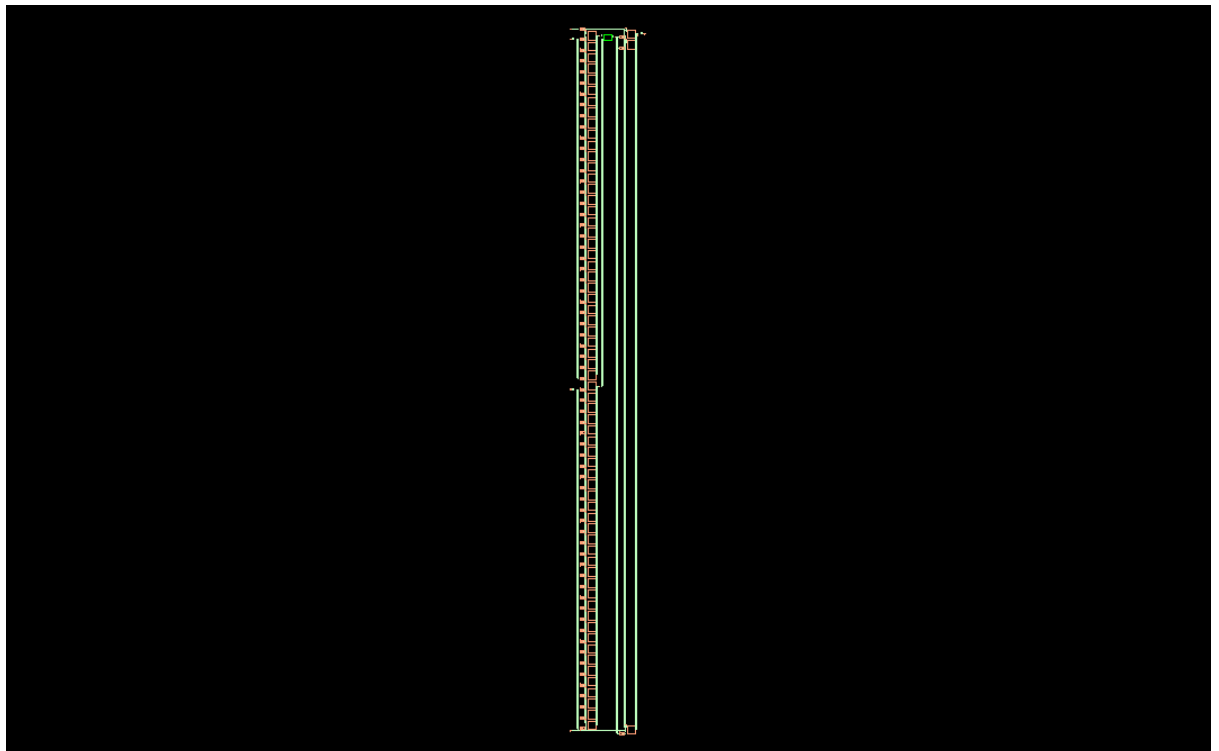
Category	Leakage	Internal	Switching	Total	Row%
memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	1.95763e-08	3.82337e-05	4.43434e-06	4.26876e-05	13.25%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	2.56924e-08	1.11745e-04	1.67669e-04	2.79440e-04	86.75%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
Subtotal	4.52688e-08	1.49978e-04	1.72104e-04	3.22127e-04	100.00%
Percentage	0.01%	46.56%	53.43%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout (fF)	Load (ps)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#	b_qd_d_reg[0]/CK	-	-	R	(arrival)	67	-	0	0	0	(-, -)
#	b_qd_d_reg[0]/Q	-	CK->Q	R	DFFRHQX1	1	5.2	101	292	292	(-, -)
	adder/g8689_6131/C0	-	A->C0	R	ADDHX1	1	5.8	111	182	473	(-, -)
	adder/g8688_7098/C0	-	CI->C0	R	ADDFX1	1	5.8	128	247	720	(-, -)
	adder/g8687_8246/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	973	(-, -)
	adder/g8686_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1225	(-, -)
	adder/g8680_5526/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1478	(-, -)
	adder/g8675_5107/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1730	(-, -)
	adder/g8671_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1983	(-, -)
	adder/g8667_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2235	(-, -)
	adder/g8660_5115/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2488	(-, -)
	adder/g8656_8246/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2740	(-, -)
	adder/g8652_1617/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2993	(-, -)
	adder/g8649_5526/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3245	(-, -)
	adder/g8636_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3498	(-, -)
	adder/g8628_7098/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3750	(-, -)
	adder/g8623_2802/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4003	(-, -)
	adder/g8619_6783/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4255	(-, -)
	adder/g8612_5477/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4508	(-, -)
	adder/g8607_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4760	(-, -)
	adder/g8603_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5013	(-, -)
	adder/g8600_4733/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5265	(-, -)
	adder/g8586_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5518	(-, -)
	adder/g8577_1666/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5770	(-, -)
	adder/g8571_6161/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6023	(-, -)
	adder/g8565_6131/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6275	(-, -)
	adder/g8555_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6528	(-, -)
	adder/g8551_2398/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6780	(-, -)
	adder/g8547_1666/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7033	(-, -)
	adder/g8544_9945/S	-	CI->S	R	ADDFX1	4	12.4	219	338	7370	(-, -)
	adder/g8541_6161/Y	-	D->Y	F	NAND4XL	3	9.7	1216	718	8089	(-, -)
	adder/g8522_6260/Y	-	B->Y	R	NAND2XL	2	6.9	341	550	8639	(-, -)
	adder/g8519_5477/Y	-	B->Y	F	NAND2BXL	2	6.9	441	345	8983	(-, -)
	adder/g8517_7410/Y	-	A0->Y	R	A0I22XL	3	9.7	558	454	9438	(-, -)
	adder/g8514_2883/Y	-	B->Y	F	NOR2XL	2	7.4	285	335	9772	(-, -)
	adder/g8512_9315/Y	-	B0->Y	F	A021X1	1	5.2	110	226	9998	(-, -)
	adder/g8491_4319/C0	-	A->C0	F	ADDHX1	3	9.4	170	171	10169	(-, -)

adder/g8491__4319/C0	-	A->C0	F	ADDHX1	3	9.4	170	171	10169	(-, -)
adder/g8487__2398/Y	-	A1->Y	R	A0I22XL	1	4.8	311	236	10405	(-, -)
adder/g8484__6417/Y	-	C0->Y	F	0AI221X1	3	9.4	552	404	10810	(-, -)
adder/g8483__7410/Y	-	B1->Y	R	A0I22XL	1	4.8	336	334	11144	(-, -)
adder/g8478__9315/Y	-	B0->Y	F	0AI211X1	3	9.4	528	409	11553	(-, -)
adder/g8470__7098/Y	-	A->Y	R	XN0R2X1	1	5.0	102	366	11919	(-, -)
adder/g8468__5122/Y	-	B->Y	F	XN0R2X1	3	9.4	180	230	12149	(-, -)
adder/g8467__1705/Y	-	B0->Y	R	A0I21XL	2	7.4	421	275	12425	(-, -)
adder/g8463__6783/Y	-	B->Y	F	XN0R2X1	1	4.8	108	302	12727	(-, -)
adder/g8461__8428/Y	-	B->Y	R	N0R2BX1	5	15.6	482	282	13010	(-, -)
adder/g8722/Y	-	A->Y	F	INVX1	2	6.9	190	246	13256	(-, -)
adder/g8457__5107/Y	-	A->Y	R	N0R2XL	2	7.4	412	278	13534	(-, -)
adder/g8453__7410/Y	-	B->Y	F	XN0R2X1	2	7.2	145	319	13853	(-, -)
adder/g8448__9315/Y	-	B->Y	R	N0R2XL	2	7.1	398	250	14104	(-, -)
adder/g8732/Y	-	A->Y	F	INVX1	2	7.9	188	222	14326	(-, -)
adder/g8443__5115/S	-	B->S	R	ADDHX1	2	7.6	136	230	14556	(-, -)
adder/g8441__6131/Y	-	B->Y	F	NAND2XL	3	9.4	575	350	14906	(-, -)
adder/g8440__7098/Y	-	A1->Y	R	A0I22XL	2	7.1	444	422	15327	(-, -)
adder/g8730/Y	-	A->Y	F	INVX1	2	7.9	196	240	15568	(-, -)
adder/g8436__1705/C0	-	B->C0	F	ADDHX1	1	4.5	91	158	15725	(-, -)
adder/g8434__2802/Y	-	B1->Y	R	0AI22XL	2	6.9	454	233	15958	(-, -)
adder/g8432__3680/Y	-	A1->Y	F	0AI21XL	1	5.0	342	349	16307	(-, -)
adder/g8430__5526/Y	-	B->Y	R	XN0R2X1	2	6.9	129	305	16612	(-, -)
adder/g8429__8428/Y	-	B0->Y	F	A0I21XL	3	10.2	636	212	16824	(-, -)
adder/g8427__6260/Y	-	B->Y	R	N0R2XL	1	4.7	325	349	17173	(-, -)
rem_qd_d_reg[2]/D	-	-	R	DFFRH0X1	1	-	-	0	17173	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area
mod_flip_flop		326	1009.926	425.808	1435.734
adder	mod5_n32	258	596.790	326.483	923.273

Genus Modulo 5 Adder µε Flip Flops 7nm

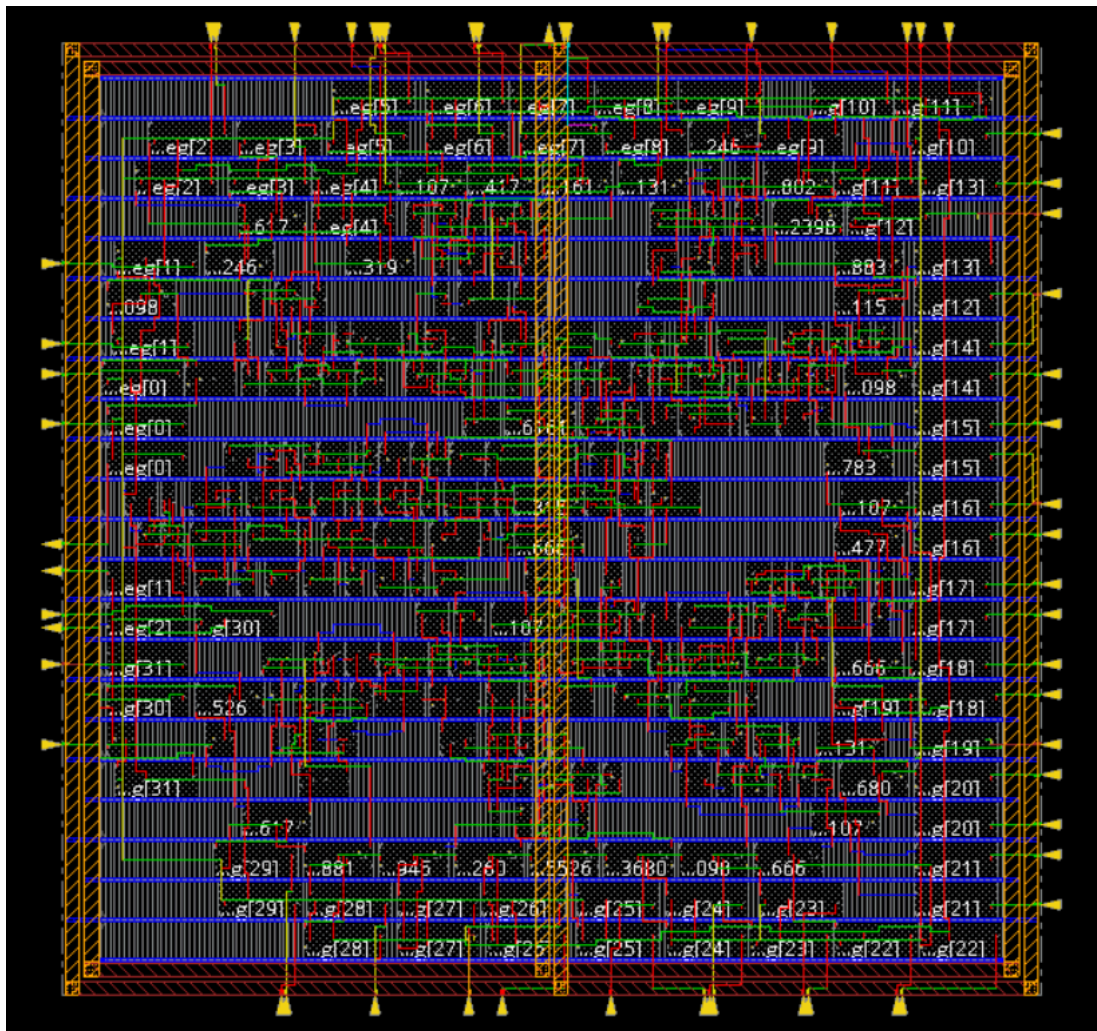


Category	Leakage	Internal	Switching	Total	Row%
memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	2.04259e-08	3.58809e-06	5.50025e-07	4.15854e-06	10.84%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	1.02869e-07	1.71215e-05	1.69686e-05	3.41930e-05	89.16%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
Subtotal	1.23295e-07	2.07096e-05	1.75186e-05	3.83515e-05	100.00%
Percentage	0.32%	54.00%	45.68%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout (fF)	Load (ps)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
	b_qd_d_reg[0]/CLK	-	-	R	(arrival)	67	-	0	0	0	(-, -)
	b_qd_d_reg[0]/QN	-	CLK->QN	F	DFFASRHQNx1_ASAP7_75t_R	2	0.6	20	50	50	(-, -)
	adder/g8939_5115/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	2	1.6	49	32	81	(-, -)
	adder/g8938/Y	-	A->Y	F	INVx1_ASAP7_75t_R	2	0.9	21	21	102	(-, -)
	adder/g8903_1705/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	25	128	(-, -)
	adder/g8902/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	144	(-, -)
	adder/g8900_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	168	(-, -)
	adder/g8898/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	185	(-, -)
	adder/g8896_5526/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	213	(-, -)
	adder/g8895/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	231	(-, -)
	adder/g8889_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	261	(-, -)
	adder/g8887/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	279	(-, -)
	adder/g8879_9315/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	308	(-, -)
	adder/g8877/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	326	(-, -)
	adder/g8872_1881/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	356	(-, -)
	adder/g8870/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	374	(-, -)
	adder/g8865_1705/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	403	(-, -)
	adder/g8862/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	422	(-, -)
	adder/g8851_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	451	(-, -)
	adder/g8849/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	469	(-, -)
	adder/g8839_4733/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	499	(-, -)
	adder/g8837/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	517	(-, -)
	adder/g8831_7098/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	546	(-, -)
	adder/g8829/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	564	(-, -)
	adder/g8824_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	594	(-, -)
	adder/g8821/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	612	(-, -)
	adder/g8812_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	641	(-, -)
	adder/g8809/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	660	(-, -)
	adder/g8791_7098/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	689	(-, -)
	adder/g8788/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	707	(-, -)
	adder/g8781_5526/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	737	(-, -)
	adder/g8779/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	755	(-, -)
	adder/g8772_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	779	(-, -)
	adder/g8769/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	796	(-, -)
	adder/g8759_5115/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	820	(-, -)
	adder/g8757/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	837	(-, -)
	adder/g8747_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	861	(-, -)

adder/g8505_8246/Y	-	B->Y	R	OR2x2_ASAP7_75t_R	1	0.5	8	19	1860	(-, -)
adder/g8502_2802/Y	-	A->Y	F	NAND4xp25_ASAP7_75t_R	2	1.4	47	24	1885	(-, -)
adder/g8501/Y	-	A->Y	R	INVxp67_ASAP7_75t_R	1	0.5	25	22	1907	(-, -)
adder/g8499_3680/Y	-	B->Y	F	NAND3xp33_ASAP7_75t_R	4	1.9	44	28	1935	(-, -)
adder/g8498/Y	-	A->Y	R	INVx1_ASAP7_75t_R	3	0.9	26	23	1957	(-, -)
adder/g8485_2346/Y	-	A->Y	F	NOR2xp33_ASAP7_75t_R	7	2.6	58	39	1997	(-, -)
adder/g8479_4733/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	2	1.3	50	42	2038	(-, -)
adder/g8470_5122/Y	-	B->Y	R	XOR2xp5_ASAP7_75t_R	2	0.7	24	31	2069	(-, -)
adder/g8466_1617/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	4	1.6	50	28	2097	(-, -)
adder/g8458_5107/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	0.6	30	24	2121	(-, -)
adder/g8456_5477/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	3	1.1	29	25	2146	(-, -)
adder/g8451_2346/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	3	1.1	38	29	2175	(-, -)
adder/g8450_2883/Y	-	B->Y	F	NAND2xp33_ASAP7_75t_R	1	0.3	16	14	2190	(-, -)
adder/g8449_9945/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	4	1.7	51	31	2221	(-, -)
adder/g8446_6161/Y	-	B->Y	R	AND2x2_ASAP7_75t_R	2	0.6	10	29	2250	(-, -)
adder/g8441_1881/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	1	0.3	15	10	2259	(-, -)
adder/g8437_8246/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	1.5	44	26	2286	(-, -)
adder/g8435_1705/Y	-	B->Y	R	AND2x2_ASAP7_75t_R	1	0.3	8	27	2313	(-, -)
adder/g8433_1617/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	1	1.5	41	21	2334	(-, -)
adder/g2/SN	-	CI->SN	R	Fx1_ASAP7_75t_R	2	0.6	37	42	2376	(-, -)
adder/g8430_5526/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	3	1.1	31	27	2403	(-, -)
adder/g8429_8428/Y	-	B->Y	F	AND2x2_ASAP7_75t_R	2	0.8	9	26	2430	(-, -)
adder/g8426_5107/Y	-	B->Y	F	OR2x2_ASAP7_75t_R	1	0.3	8	21	2451	(-, -)
g14/Y	-	A->Y	R	INVxp33_ASAP7_75t_R	1	0.6	31	19	2470	(-, -)
rem_qd_d_reg[0]/D	-	-	R	DFASRHQNX1_ASAP7_75t_R	1	-	-	0	2470	(-, -)

Innovus Modulo 5 Adder μ e Flip Flops 45nm



```
# Generated by: Cadence Innovus 22.33-s094_1
# OS: Linux x86_64(Host ID r540)
# Generated on: Tue Jul 8 09:46:32 2025
# Design: mod_flip_flop
# Command: report timing
```

```
#####
Path 1: MET Setup Check with Pin rem_qd_d_reg[0]/CK
Endpoint:  rem_qd_d_reg[0]/D (^) checked with  leading edge of 'clk'
Beginpoint: b_qd_d_reg[0]/Q (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
```

```
Analysis View: worst_case
Other End Arrival Time      -0.000
- Setup                     0.109
+ Phase Shift               18.000
= Required Time             17.890
- Arrival Time              11.106
= Slack Time                 6.785

    Clock Rise Edge          0.000
    + Clock Network Latency (Prop) -0.000
    = Beginpoint Arrival Time -0.000
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
b_qd_d_reg[0]	CK ^			-0.000	6.784
b_qd_d_reg[0]	CK ^ -> Q ^	DFFRHQX1	0.271	0.271	7.055
adder/g7459_6131	A ^ -> C0 ^	ADDFX1	0.143	0.414	7.198
adder/g7458_7098	CI ^ -> C0 ^	ADDFX1	0.209	0.623	7.407
adder/g7457_8246	CI ^ -> C0 ^	ADDFX1	0.213	0.836	7.621
adder/g7453_1617	CI ^ -> C0 ^	ADDFX1	0.212	1.048	7.833
adder/g7448_4319	CI ^ -> C0 ^	ADDFX1	0.214	1.262	8.047
adder/g7446_5107	CI ^ -> C0 ^	ADDFX1	0.213	1.475	8.260
adder/g7443_6417	CI ^ -> C0 ^	ADDFX1	0.212	1.687	8.472
adder/g7435_6161	CI ^ -> C0 ^	ADDFX1	0.214	1.901	8.686
adder/g7430_6131	CI ^ -> C0 ^	ADDFX1	0.213	2.114	8.898
adder/g7428_8246	CI ^ -> C0 ^	ADDFX1	0.212	2.326	9.111
adder/g7425_2802	CI ^ -> C0 ^	ADDFX1	0.212	2.538	9.323
adder/g7415_2398	CI ^ -> C0 ^	ADDFX1	0.211	2.750	9.534

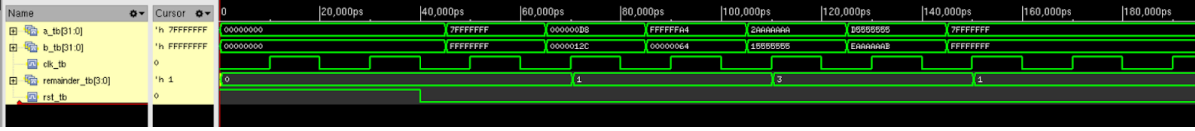
adder/g7415	2398	CI ^ -> C0 ^	ADDFX1	0.211	2.750	9.534
adder/g7408	2883	CI ^ -> C0 ^	ADDFX1	0.210	2.959	9.744
adder/g7402	5115	CI ^ -> C0 ^	ADDFX1	0.210	3.169	9.954
adder/g7399	7098	CI ^ -> C0 ^	ADDFX1	0.210	3.379	10.164
adder/g7391	6783	CI ^ -> C0 ^	ADDFX1	0.212	3.591	10.376
adder/g7386	5107	CI ^ -> C0 ^	ADDFX1	0.210	3.801	10.586
adder/g7384	5477	CI ^ -> C0 ^	ADDFX1	0.210	4.012	10.796
adder/g7381	1666	CI ^ -> C0 ^	ADDFX1	0.209	4.220	11.005
adder/g7371	6131	CI ^ -> C0 ^	ADDFX1	0.210	4.430	11.215
adder/g7362	3680	CI ^ -> C0 ^	ADDFX1	0.210	4.640	11.425
adder/g7356	5107	CI ^ -> C0 ^	ADDFX1	0.209	4.849	11.633
adder/g7351	1666	CI ^ -> C0 ^	ADDFX1	0.208	5.057	11.841
adder/g7340	7098	CI ^ -> C0 ^	ADDFX1	0.208	5.265	12.050
adder/g7333	3680	CI ^ -> C0 ^	ADDFX1	0.208	5.473	12.258
adder/g7331	5526	CI ^ -> C0 ^	ADDFX1	0.208	5.682	12.466
adder/g7328	6260	CI ^ -> C0 ^	ADDFX1	0.208	5.889	12.674
adder/g7319	9945	CI ^ -> C0 ^	ADDFX1	0.208	6.098	12.882
adder/g7310	1881	CI ^ -> C0 ^	ADDFX1	0.208	6.306	13.090
adder/g7303	1617	CI ^ -> C0 ^	ADDFX1	0.210	6.516	13.301
adder/g7300	5526	CI ^ -> C0 ^	ADDFX1	0.204	6.720	13.505
adder/g7296	5107	B ^ -> Y v	XNOR2X1	0.179	6.899	13.683
adder/g7468		A v -> Y ^	INVX1	0.069	6.968	13.752
adder/g7470	5115	C ^ -> Y v	NAND3BXL	0.157	7.125	13.909
adder/g7285	9315	A v -> Y v	AND2X1	0.169	7.294	14.078
adder/g7282	7482	B v -> C0 v	ADDHX1	0.085	7.379	14.163
adder/g7280	1881	B v -> Y ^	XNOR2X1	0.188	7.567	14.351
adder/g7277	8246	B ^ -> Y v	XNOR2X1	0.153	7.720	14.505
adder/g7274	2802	B v -> Y ^	XNOR2X1	0.186	7.906	14.690
adder/g7265	5107	CI ^ -> S v	ADDFX1	0.305	8.210	14.995
adder/g7260	1666	CI v -> C0 v	ADDFX1	0.204	8.415	15.199
adder/g7256	9315	CI v -> C0 v	ADDFX1	0.207	8.621	15.406
adder/g7254	4733	B v -> Y ^	XNOR2X1	0.204	8.825	15.610
adder/g7249	6131	A ^ -> C0 ^	ADDHX1	0.142	8.967	15.751
adder/g7246	5122	B ^ -> Y ^	CLKXOR2X1	0.139	9.105	15.890
adder/g7244	2802	B ^ -> Y ^	AND2X1	0.145	9.250	16.035
adder/g7242	3680	AN ^ -> Y ^	NOR2BX1	0.132	9.382	16.166
adder/g7240	5526	B ^ -> Y v	XNOR2X1	0.167	9.549	16.333
adder/g7237	6260	B v -> Y ^	NOR2X1	0.068	9.617	16.402
adder/g7490		A ^ -> Y v	INVX1	0.067	9.684	16.469
adder/g7229	2346	B v -> C0 v	ADDHX1	0.085	9.769	16.554
adder/g7228	2883	B v -> Y ^	XNOR2X1	0.178	9.947	16.732
adder/g7224	4733	B ^ -> Y v	NAND2X1	0.079	10.026	16.810

adder/g7228_2883	B v -> Y ^	XNOR2X1	0.178	9.947	16.732
adder/g7224_4733	B ^ -> Y ^	NAND2X1	0.079	10.026	16.810
adder/g7223_7482	B v -> Y ^	NAND2BX1	0.065	10.090	16.875
adder/g7222_5115	B ^ -> Y v	NAND2X1	0.108	10.198	16.983
adder/g7217_5122	B v -> C0 v	ADDHX1	0.122	10.321	17.105
adder/g7216_1705	B v -> Y ^	XNOR2X1	0.175	10.495	17.280
adder/g7215_2802	AN ^ -> Y ^	NAND2BX1	0.080	10.575	17.360
adder/g7213_3680	B ^ -> Y v	NAND2X1	0.056	10.631	17.416
adder/g7211_5526	B v -> Y ^	XNOR2X1	0.194	10.825	17.609
adder/g7210_8428	B0 ^ -> Y v	A0I21X1	0.065	10.890	17.674
adder/g7207_5107	B v -> Y ^	XNOR2X1	0.216	11.106	17.890
rem_qd_d_reg[0]	D ^	DFFRHQX1	0.000	11.106	17.890

LEC για Modulo 5 adder με Flip Flops

Verification Report	
Category	Count
1. Non-standard modeling options used:	0
2. Incomplete verification:	0
3. User modification to design:	0
4. Conformal Constraint Designer clock domain crossing checks recommended:	0
5. Design ambiguity:	0
6. Compare Results:	PASS

Xcelium για Modulo 5 adder με Flip Flops



Modulo 9 Adder

Τα components για τον αθροιστή και τα registers είναι ίδια.

Συγκριτής για το mod9

```
library ieee; use ieee.std_logic_1164.all;

entity comparator_of_9 is generic (n:integer:=6); port(
    S: in std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
);
end comparator_of_9;

architecture dataflow of comparator_of_9 is

    signal above_below : std_logic_vector(1 downto 0);
begin
    above_below<="10" when ((S(5)='1'or S(4)='1') and negative='0') or
(negative='0' and S(3) = '1' and (S(2)='1' or S(1) = '1' or S(0) = '1'))
else
        "01" when (S(n-1) = '1' and negative='1') else
        "00";

    above <= above_below(1);
    below <= above_below(0);
end dataflow;
```

/comparator_of_9/n	6	6					
/comparator_of_9/S	101000	001001		001000		101000	
/comparator_of_9/negative	1						
/comparator_of_9/above	0						
/comparator_of_9/below	1						
/comparator_of_9/above_bel...	01	10		00		01	

Υπολογισμός κρατουμένου για το Chunk

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity remainder_calculator_9 is
    port (
        data_in      : in  std_logic_vector(5 downto 0);
        negative_first: in  std_logic;
        remainder     : out std_logic_vector(5 downto 0)
    );
end entity;

architecture rtl of remainder_calculator_9 is

    signal mod9, mod9_comp : std_logic_vector(5 downto 0);

    type slv4_array is array (natural range <>) of std_logic_vector(5
downto 0);
    signal data : slv4_array(0 to 6);

    component comparator_of_9 is generic (n:integer:=6); port(
        S: in  std_logic_vector(n-1 downto 0);
        negative: in std_logic;
        above: out std_logic;
        below: out std_logic
    );
end component;

    component adder_n is
        generic (n: integer := 6);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
        );
    end component;

begin
    mod9 <= std_logic_vector(to_signed(9, 6));
    mod9_comp <= std_logic_vector(to_signed(-9, 6));
```

```

    data(0) <= "000000" when (data_in = "111111" and negative_first = '0')
else --check for last one
    data_in;

generate_label:
for i in 0 to 5 generate
    signal above_i, below_i: std_logic;
    signal temp1_i, temp2_i: std_logic_vector(5 downto 0);
begin
    -- First comparison stage
    comp_i: comparator_of_9
        generic map(6)
        port map(data(i), negative_first, above_i, below_i);

    -- First arithmetic stage
    add_above_i: adder_n
        generic map(6)
        port map(data(i), mod9_comp, '0', temp1_i, open);

    add_below_i: adder_n
        generic map(6)
        port map(data(i), mod9, '0', temp2_i, open);

    -- Intermediate mux
    data(i+1) <= temp1_i when above_i = '1' else
        temp2_i when below_i = '1' else
        data(i);

end generate;
-- Final output selection
remainder <= data(6);

end architecture;

```

 /remainder_calculator_9/data_in	-32	9		8		-32	
 /remainder_calculator_9/negative_first	1						
 /remainder_calculator_9/remainder	4	0		8		4	
 /remainder_calculator_9/mod9	9	9					
 /remainder_calculator_9/mod9_comp	-9	-9					
 /remainder_calculator_9/data	{-32} ...	{9} {0} {0} {...	{8} {8} {8} {...	{-32} {-23} {-...			

Υπολογισμός mod 9

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity mod9 is
    generic (n: integer := 32);
    port(
        A, B: in std_logic_vector(n-1 downto 0);
        remainder: out std_logic_vector(5 downto 0) -- [5-0] signed
    );
end mod9;

architecture dataflow of mod9 is

    signal S: std_logic_vector(n-1 downto 0);
    signal padded_36 : std_logic_vector(35 downto 0);
    signal Module, Module_Complement : std_logic_vector(5 downto 0);

    -- Correct array type declaration
    type slv6_array is array (natural range <>) of std_logic_vector(5
downto 0);
    signal remainders : slv6_array(0 to 5);

    -- Adder tree signals
    signal sum_stage1 : slv6_array(0 to 2);
    signal sum_stage2 : slv6_array(0 to 0);

    signal sum_final : std_logic_vector(5 downto 0);
    signal sum_final_temp1, sum_final_temp2: std_logic_vector(5 downto 0);
    signal abovesum, belowsum: std_logic;

    -- decide if i want to calculate as if its a negative or positive
number
    signal negative: std_logic;
    signal s_first: std_logic_vector(5 downto 0);
    -- Component declarations
    component adder_n is
        generic (n: integer := 6);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
```

```

    );
end component;

component comparator_of_9 is
    generic (n: integer := 6);
    port(
        S: in std_logic_vector(n-1 downto 0);
        negative: in std_logic;
        above: out std_logic;
        below: out std_logic
    );
end component;

component remainder_calculator_9 is
port (
    data_in      : in  std_logic_vector(5 downto 0);
    negative_first: in  std_logic;
    remainder     : out std_logic_vector(5 downto 0)
);
end component;

begin
Module <= std_logic_vector(to_signed(9, 6));
Module_Complement <= std_logic_vector(to_signed(-9, 6));

adder : adder_n
    generic map(n)
    port map (A, B, '0', S, open);

padded_36 <= (35 downto n => S(n-1)) & S;

negative <= padded_36(35);

s_first <= padded_36(35 downto 30);

remainder_c : remainder_calculator_9
port map(s_first,negative,
remainders(5));

generate_label:
for i in 0 to 4 generate

signal s_in: std_logic_vector(5 downto 0);
begin

```

```

s_in <= padded_36(5+i*6 downto i*6);

remainder_c : remainder_calculator_9
port map(s_in,'0',
remainders(i));
end generate;

-- Adder tree to sum all remainders
-- Stage 1: Add pairs of remainders

gen_stage1: for i in 0 to 2 generate
  adder_st1: adder_n
    generic map(6)
    port map(
      a => remainders(i*2),
      b => remainders(i*2+1),
      cin => '0',
      sum => sum_stage1(i),
      cout => open
    );
end generate gen_stage1;

-- Stage 2: Add results from stage 1

  adder_st2: adder_n
    generic map(6)
    port map(
      a => sum_stage1(0),
      b => sum_stage1(1),
      cin => '0',
      sum => sum_stage2(0),
      cout => open
    );

-- Final adder
adder_final: adder_n
  generic map(6)
  port map(
    a => sum_stage1(2),
    b => sum_stage2(0),
    cin => '0',
    sum => sum_final,
    cout => open
  );

```

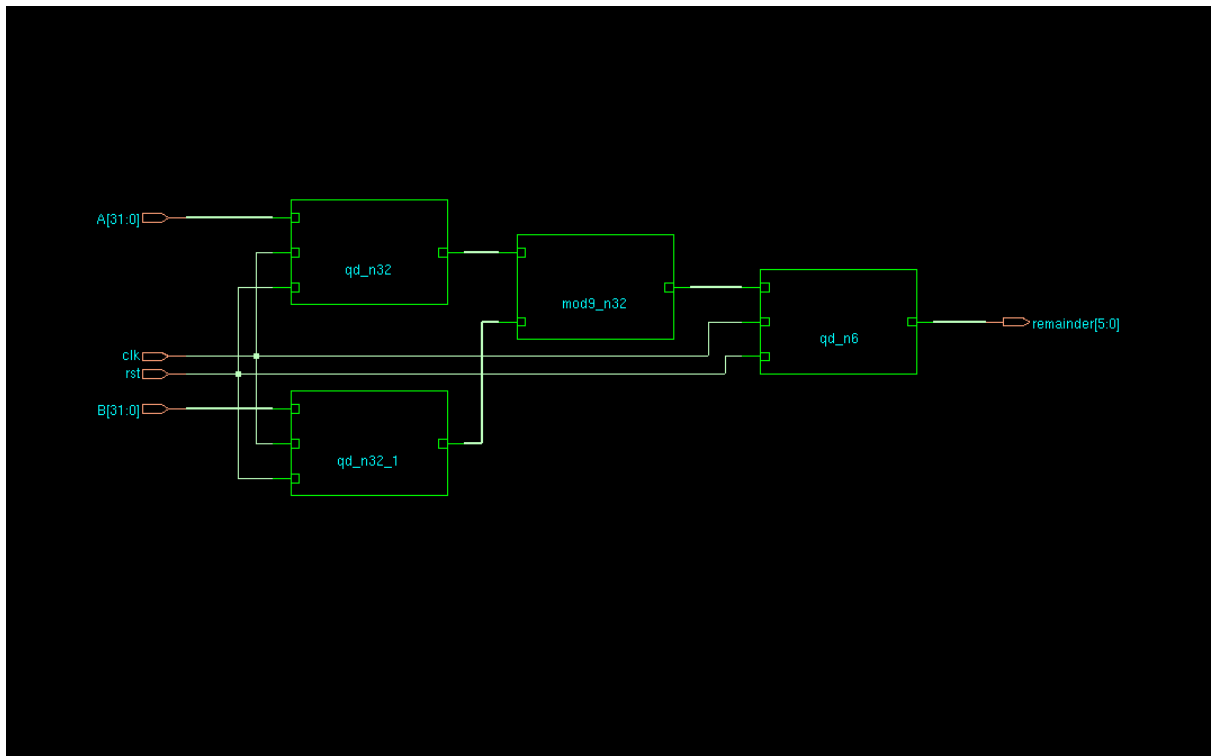
```
remainder_sum : remainder_calculator_9
port map(sum_final, '0',
remainder);
```

 /mod9/n	32	32
 /mod9/A	136348168	136348168
 /mod9/B	0	0
 /mod9/remainder	4	4
 /mod9/S	136348168	136348168
 /mod9/padded_36	0000000...	000000001000001000001000001000001000
 /mod9/Module	9	9
 /mod9/Module_Complement	-9	-9
 /mod9/remainders	{001000}...	{001000} {001000} {001000} {001000} {001000} {000000}
 /mod9/sum_stage1	{16} {16...	{16} {16} {8}
 /mod9/sum_stage2	{32}	{32}
 /mod9/sum final	40	40

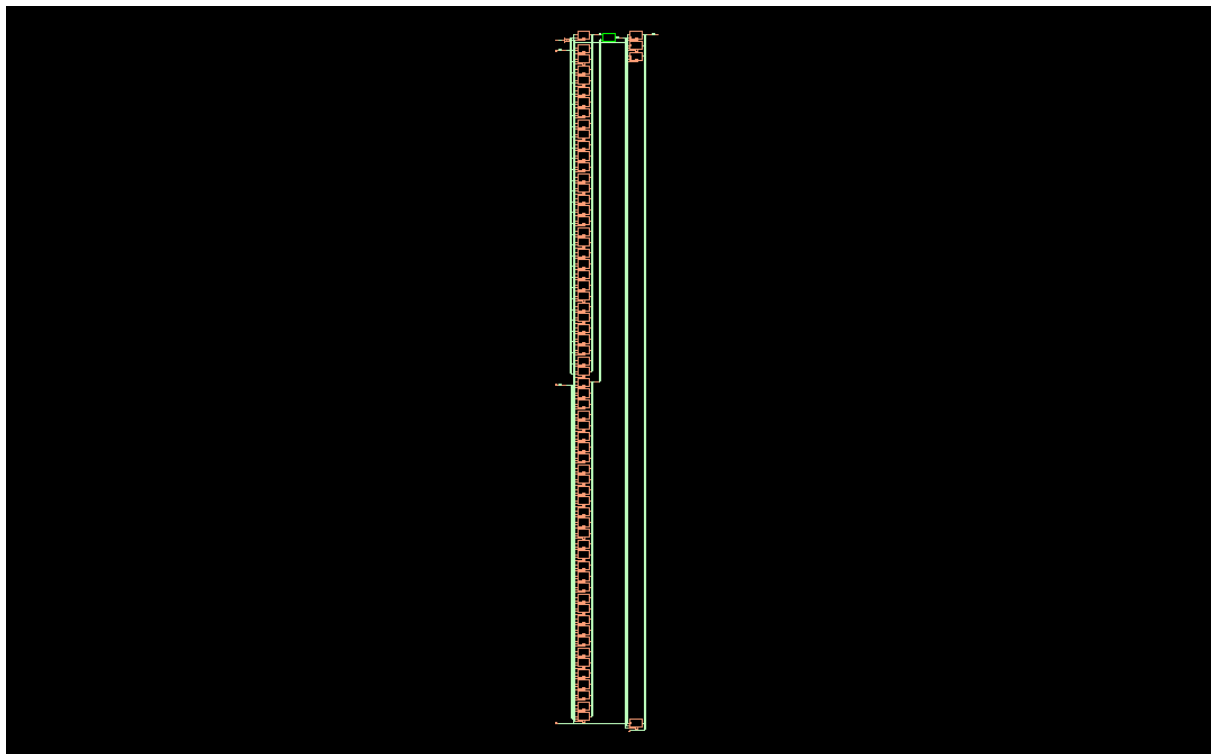
Testbench με flip flops

◆ /testbenc...	2147483647	216		-92	715827882	-715827883	2147483647	
◆ /testbenc...	-1	300		100	357913941	-357913941	-1	
◆ /testbenc...	1							
◆ /testbenc...	0							
◆ /testbenc...	0	X	3	8	0	8	0	

Genus Modulo 9 Adder με Flip Flops



Genus Modulo 9 Adder με Flip Flops 45nm



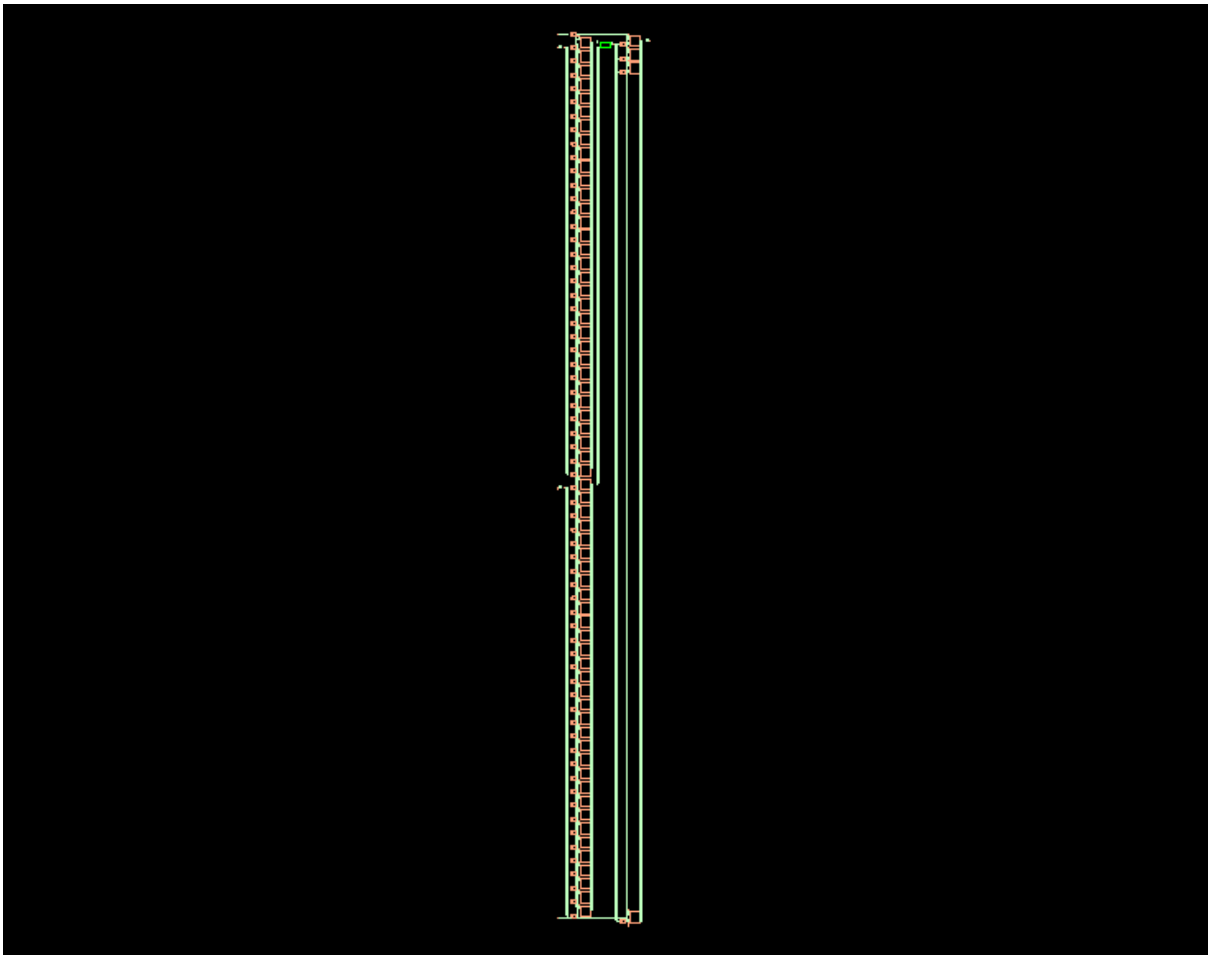
Category	Leakage	Internal	Switching	Total	Row%
memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	1.98685e-08	6.14438e-05	5.06417e-06	6.65278e-05	44.50%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	2.80212e-08	3.25104e-05	4.78807e-05	8.04191e-05	53.80%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	2.54215e-06	2.54215e-06	1.70%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
Subtotal	4.78897e-08	9.39541e-05	5.54870e-05	1.49489e-04	100.00%
Percentage	0.03%	62.85%	37.12%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
#											
	b_qd_d_reg[0]/CK	-	-	R	(arrival)	68	-	0	0	0	(-, -)
	b_qd_d_reg[0]/Q	-	CK->Q	R	DDFRHQX1	1	5.2	101	292	292	(-, -)
	adder/g5708_5526/C0	-	A->C0	R	ADDHX1	1	5.8	111	182	473	(-, -)
	adder/g5707_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	128	247	720	(-, -)
	adder/g5706_4319/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	973	(-, -)
	adder/g5705_6260/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1225	(-, -)
	adder/g5702_2398/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1478	(-, -)
	adder/g5696_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1730	(-, -)
	adder/g5686_5115/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1983	(-, -)
	adder/g5674_6783/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2235	(-, -)
	adder/g5665_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2488	(-, -)
	adder/g5663_7410/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2740	(-, -)
	adder/g5658_9945/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2993	(-, -)
	adder/g5653_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3245	(-, -)
	adder/g5642_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3498	(-, -)
	adder/g5634_4319/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3750	(-, -)
	adder/g5626_7410/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4003	(-, -)
	adder/g5621_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4255	(-, -)
	adder/g5617_7482/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4508	(-, -)
	adder/g5607_2802/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4760	(-, -)
	adder/g5597_6260/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5013	(-, -)
	adder/g5585_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5265	(-, -)
	adder/g5577_8246/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5518	(-, -)
	adder/g5576_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5770	(-, -)
	adder/g5572_1617/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6023	(-, -)
	adder/g5567_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6275	(-, -)
	adder/g5557_7410/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6528	(-, -)
	adder/g5545_1881/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6780	(-, -)
	adder/g5535_6783/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7033	(-, -)
	adder/g5531_6260/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7285	(-, -)
	adder/g5527_5477/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7538	(-, -)
	adder/g5517_6161/S	-	CI->S	R	ADDFX1	3	10.3	190	322	7859	(-, -)
	adder/g5713_3680/Y	-	AN->Y	R	NOR2BX1	4	13.1	412	322	8181	(-, -)
	adder/g5494_6260/Y	-	B->Y	R	XNOR2X1	1	4.7	98	275	8456	(-, -)
	adder/g5479_7482/Y	-	B1->Y	F	OAI222X1	1	4.8	370	263	8719	(-, -)
	adder/g5476_1881/Y	-	B0->Y	R	AOI21X1	1	4.8	209	214	8933	(-, -)
	adder/g5472_5122/Y	-	B0->Y	F	OAI2BB1X1	1	6.3	234	202	9135	(-, -)

adder/g5479__7482/Y	-	B1->Y	F	0AI222X1	1	4.8	370	263	8719	(-, -)
adder/g5476__1881/Y	-	B0->Y	R	A0I21X1	1	4.8	209	214	8933	(-, -)
adder/g5472__5122/Y	-	B0->Y	F	0AI2BB1X1	1	6.3	234	202	9135	(-, -)
adder/g5469__1617/C0	-	B->C0	F	ADDFX1	2	7.8	172	308	9443	(-, -)
adder/g5467__6783/Y	-	A1->Y	R	A0I22X1	1	4.8	197	176	9619	(-, -)
adder/g5466__5526/Y	-	C0->Y	F	0AI221X1	1	6.3	408	292	9912	(-, -)
adder/g5463__6260/S	-	B->S	R	ADDFX1	6	18.5	305	542	10454	(-, -)
adder/g5807/Y	-	A->Y	F	INVX1	2	6.9	156	179	10633	(-, -)
adder/g5459__5477/Y	-	B->Y	F	AND2X1	6	17.6	302	250	10883	(-, -)
adder/g5445__5115/Y	-	A0->Y	R	A0I22X1	1	4.7	214	221	11104	(-, -)
adder/g5435__3680/Y	-	A1->Y	F	0AI22X1	1	4.8	231	206	11310	(-, -)
adder/g5430__6260/Y	-	B0->Y	R	A0I21X1	1	4.7	186	167	11476	(-, -)
adder/g5428__2398/Y	-	C0->Y	F	0AI211X1	1	4.7	307	231	11707	(-, -)
rem_qd_d_reg[1]/D	<<<	-	F	DFFRHQX1	1	-	-	0	11707	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area
mod_flip_flop		349	1036.944	460.950	1497.894
adder	mod9_n32	280	617.652	360.142	977.794

Genus Modulo 9 Adder μ e Flip Flops 7nm



Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	2.07228e-08	2.17029e-06	5.44145e-07	2.73515e-06	23.81%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	9.97904e-08	3.84991e-06	4.80487e-06	8.75457e-06	76.19%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	1.20513e-07	6.02020e-06	5.34901e-06	1.14897e-05	100.00%
Percentage	1.05%	52.40%	46.55%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
#											
	a_qd_d_reg[0]/CLK	-	-	R	(arrival)	68	-	0	0	0	(-, -)
	a_qd_d_reg[0]/QN	-	CLK->QN	F	DFFASRHQNx1_ASAP7_75t_R	1	1.0	24	53	53	(-, -)
	adder/g6368/CON	-	A->CON	R	HAXp5_ASAP7_75t_R	2	1.6	35	25	78	(-, -)
	adder/g6335/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	16	94	(-, -)
	adder/g6302_1666/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	28	122	(-, -)
	adder/g6301/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	140	(-, -)
	adder/g6299_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	165	(-, -)
	adder/g6298/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	182	(-, -)
	adder/g6295_9315/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	210	(-, -)
	adder/g6294/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	229	(-, -)
	adder/g6288_5115/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	253	(-, -)
	adder/g6286/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	270	(-, -)
	adder/g6278_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	294	(-, -)
	adder/g6277/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	310	(-, -)
	adder/g6264_7410/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	334	(-, -)
	adder/g6258/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	351	(-, -)
	adder/g6241_8428/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	379	(-, -)
	adder/g6239/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	398	(-, -)
	adder/g6233_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	422	(-, -)
	adder/g6232/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	439	(-, -)
	adder/g6226_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	468	(-, -)
	adder/g6225/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	486	(-, -)
	adder/g6218_7482/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	510	(-, -)
	adder/g6217/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	527	(-, -)
	adder/g6209_1705/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	551	(-, -)
	adder/g6208/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	567	(-, -)
	adder/g6195_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	591	(-, -)
	adder/g6189/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	608	(-, -)
	adder/g6172_6783/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	632	(-, -)
	adder/g6170/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	648	(-, -)
	adder/g6160_7410/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	677	(-, -)
	adder/g6159/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	695	(-, -)
	adder/g6153_6161/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	724	(-, -)
	adder/g6150/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	742	(-, -)
	adder/g6141_5122/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	772	(-, -)
	adder/g6140/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	790	(-, -)
	adder/g6126_2398/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	814	(-, -)
	adder/g6124/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	831	(-, -)
	adder/g6104/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	872	(-, -)
	adder/g6087_2346/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	900	(-, -)
	adder/g6085/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	919	(-, -)
	adder/g6079_7482/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	948	(-, -)
	adder/g6078/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	966	(-, -)
	adder/g6073_7098/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	995	(-, -)
	adder/g6072/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	1014	(-, -)
	adder/g6065_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	1043	(-, -)
	adder/g6064/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	1061	(-, -)
	adder/g6055_5107/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	1086	(-, -)
	adder/g6053/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1103	(-, -)
	adder/g6039_7482/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	1127	(-, -)
	adder/g6034/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	1143	(-, -)
	adder/g6017_1666/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	1172	(-, -)
	adder/g6015/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	1190	(-, -)
	adder/g6004_6131/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	1219	(-, -)
	adder/g6002/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	1237	(-, -)
	adder/g5994_6783/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	49	29	1267	(-, -)
	adder/g5991/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	19	18	1285	(-, -)
	adder/g5980_2346/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	1310	(-, -)
	adder/g5979/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1326	(-, -)
	adder/g5966_8246/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	1355	(-, -)
	adder/g5965/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	1373	(-, -)
	adder/g5948_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	1	1.5	39	24	1398	(-, -)
	adder/g6347/SN	-	CI->SN	F	FAX1_ASAP7_75t_R	3	0.9	33	50	1448	(-, -)
	adder/g5912_2346/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	2	1.2	40	30	1479	(-, -)
	adder/g5910/Y	-	A->Y	F	INVx1_ASAP7_75t_R	4	4.2	48	38	1517	(-, -)
	adder/g5904_7482/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	1.3	45	34	1551	(-, -)
	adder/g5897_5122/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	2	1.6	42	35	1586	(-, -)
	adder/g5888_4319/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.5	60	40	1626	(-, -)
	adder/g6346/SN	-	CI->SN	F	FAX1_ASAP7_75t_R	2	2.5	66	87	1714	(-, -)
	adder/g5879_2883/Y	-	C->Y	F	MAJx2_ASAP7_75t_R	2	2.1	15	44	1757	(-, -)
	adder/g5874_6161/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	2	1.8	14	30	1787	(-, -)
	adder/g2/SN	-	CI->SN	R	FAX1_ASAP7_75t_R	4	1.9	71	58	1845	(-, -)
	adder/g5863_5122/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	6	2.6	65	54	1899	(-, -)
	adder/g5861/Y	-	A->Y	R	INVxp67_ASAP7_75t_R	1	0.3	26	21	1921	(-, -)
	adder/g5842_9945/Y	-	A->Y	F	NOR2xp33_ASAP7_75t_R	2	0.8	26	21	1942	(-, -)
	adder/g5837_5115/Y	-	B->Y	F	OR2x2_ASAP7_75t_R	1	0.3	8	26	1968	(-, -)
	adder/g5828_3680/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	1	0.5	26	13	1981	(-, -)
	adder/g5822_5107/Y	-	A->Y	F	NAND4xp25_ASAP7_75t_R	1	0.3	24	17	1998	(-, -)
	g13/Y	-	A->Y	R	INVxp33_ASAP7_75t_R	1	0.6	33	25	2023	(-, -)
	rem_qd_d_reg[0]/D	-	-	R	DFFASRHQNx1_ASAP7_75t_R	1	-	-	0	2023	(-, -)

LEC Modulo 9 Adder με Flip Flops

Verification Report	
Category	Count
1. Non-standard modeling options used:	0
2. Incomplete verification:	0
3. User modification to design:	0
4. Conformal Constraint Designer clock domain crossing checks recommended:	0
5. Design ambiguity:	0
6. Compare Results:	PASS

Xcelium Modulo 9 Adder με Flip Flops



Modulo 14 Adder

Δημιουργία mod7

Συγκριτής 7

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity comparator_of_7 is
  generic (n: integer := 3);
  port (
    S: in std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
  );
end comparator_of_7;

architecture dataflow of comparator_of_7 is

  signal above_below: std_logic_vector(1 downto 0);
  signal S_padded : std_logic_vector(6 downto 0); -- always 7 bits
begin
  S_padded <= (6 downto n => '0') & S;

  above_below <= "10" when (
    (S_padded(6 downto 3) /= "0000" and negative = '0') or
    (negative = '0' and S(2) = '1' and S(1) = '1' and S(0)
= '1')
    ) else
    "01" when (S(n-1) = '1' and negative = '1') else
    "00";

  above <= above_below(1);
  below <= above_below(0);
end dataflow;
```

Υπολογισμός υπολοίπου mod7 για μικρό τμήμα

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity remainder_calculator_7 is generic (n:integer:=7);
  port (
    data_in      : in  std_logic_vector(n-1 downto 0);
    negative_first: in  std_logic;
    remainder     : out std_logic_vector(n-1 downto 0)
  );
end entity;

architecture rtl of remainder_calculator_7 is

  signal mod7, mod7_comp : std_logic_vector(6 downto 0);
  constant all_ones : std_logic_vector(n-1 downto 0) := (others =>
'1');

  type slv6_array is array (natural range <>) of std_logic_vector(6
downto 0);
  signal data : slv6_array(0 to 10);

  component comparator_of_7 is generic (n:integer:=4); port(
    S: in  std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
  );
end component;

  component adder_n is
    generic (n: integer := 4);
    port(
      a, b: in std_logic_vector(n-1 downto 0);
      cin: in std_logic;
      sum: out std_logic_vector(n-1 downto 0);
      cout: out std_logic
    );
  end component;

begin
  mod7 <= std_logic_vector(to_signed(7, 7));
  mod7_comp <= std_logic_vector(to_signed(-7, 7));
```

```

data(0) <= (others => '0') when (data_in = all_ones and negative_first =
'0') else
    (6 downto n => data_in(n-1)) & data_in when negative_first
='1' else
    (6 downto n => '0') & data_in;

generate_label:
for i in 0 to 9 generate
    signal above_i, below_i: std_logic;
    signal temp1_i, temp2_i: std_logic_vector(6 downto 0);
begin
    -- First comparison stage
    comp_i: comparator_of_7
        generic map(7)
        port map(data(i), negative_first, above_i, below_i);

    -- First arithmetic stage
    add_above_i: adder_n
        generic map(7)
        port map(data(i), mod7_comp, '0', temp1_i, open);

    add_below_i: adder_n
        generic map(7)
        port map(data(i), mod7, '0', temp2_i, open);

    -- Intermediate mux
    data(i+1) <= temp1_i when above_i = '1' else
        temp2_i when below_i = '1' else
        data(i);

end generate;

-- Final output selection
remainder<=data(10)(n-1 downto 0);

end architecture;

```

Συνολικός Υπολογισμός mod7

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity mod7 is
    generic (n: integer := 32);
    port(
        A, B: in std_logic_vector(n-1 downto 0);
        remainder: out std_logic_vector(3 downto 0) -- [4-0] signed
    );
end mod7;

architecture dataflow of mod7 is

    signal S: std_logic_vector(n-1 downto 0);
    signal padded_33 : std_logic_vector(32 downto 0);

    -- Correct array type declaration
    type slv4_array is array (natural range <>) of std_logic_vector(2
downto 0);
    signal remainders : slv4_array(0 to 10);
    signal sum_remain : std_logic_vector(6 downto 0);
    type slv7_array is array (natural range <>) of std_logic_vector(6
downto 0);
    -- Adder tree signals
    signal sum_stage1 : slv7_array(0 to 5);
    signal sum_stage2 : slv7_array(0 to 2);
    signal sum_final,sum_final_temp: std_logic_vector(6 downto 0);

    -- decide if i want to calculate as if its a negative or positive
number
    signal negative:std_logic;
    signal s_first:std_logic_vector(2 downto 0);
    -- Component declarations
    component adder_n is
        generic (n: integer := 4);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
        );
    end component;
```

```

component comparator_of_7 is
    generic (n: integer := 4);
    port(
        S: in std_logic_vector(n-1 downto 0);
        negative: in std_logic;
        above: out std_logic;
        below: out std_logic
    );
end component;

component remainder_calculator_7 is generic (n: integer := 4);
port (
    data_in      : in  std_logic_vector(n-1 downto 0);
    negative_first: in  std_logic;
    remainder     : out std_logic_vector(n-1 downto 0)
);
end component;

begin

adder : adder_n
    generic map(n)
    port map (A, B, '0', S, open);

padded_33 <= (32 downto n => S(n-1)) & S;

negative <= padded_33(32);

s_first <= padded_33(32 downto 30);

remainder_c : remainder_calculator_7
    generic map(3)
    port map(s_first, negative,
        remainders(10));

generate_label:
for i in 0 to 9 generate

    signal s_in: std_logic_vector(2 downto 0);
    begin

        s_in <= padded_33(2+i*3 downto i*3);

```

```

remainder_c : remainder_calculator_7
generic map(3)
port map(s_in,'0',
remainders(i));
end generate;

-- Adder tree to sum all remainders
-- Stage 1: Add pairs of remainders

gen_stage1: for i in 0 to 4 generate

    signal a_padded, b_padded : std_logic_vector(6 downto 0);
begin
    a_padded <= "0000" & remainders(i*2);
    b_padded <= "0000" & remainders(i*2+1);

    adder_st1: adder_n
        generic map(7)
        port map(
            a => a_padded,
            b => b_padded,
            cin => '0',
            sum => sum_stage1(i),
            cout => open
        );
end generate gen_stage1;

sum_stage1(5)<="0000"&remainders(10);

-- Stage 2: Add results from stage 1
gen_stage2: for i in 0 to 2 generate
    adder_st2: adder_n
        generic map(7)
        port map(
            a => sum_stage1(i*2),
            b => sum_stage1(i*2+1),
            cin => '0',
            sum => sum_stage2(i),
            cout => open
        );
end generate gen_stage2;

-- Pre Final adder
adder_prefinal: adder_n

```

```

    generic map(7)
    port map(
        a => sum_stage2(0),
        b => sum_stage2(1),
        cin => '0',
        sum => sum_final_temp,
        cout => open
    );
-- Final adder
adder_final: adder_n
    generic map(7)
    port map(
        a => sum_stage2(2),
        b => sum_final_temp,
        cin => '0',
        sum => sum_final,
        cout => open
    );

remainder_sum : remainder_calculator_7
    generic map(7)
    port map(sum_final,'0',
sum_remain);

remainder <= sum_remain(3 downto 0);

end dataflow;

```

	Msgs	
/mod7/n	32	32
/mod7/A	1101010...	11010101010101010101010101010101
/mod7/B	1110101...	11101010101010101010101010101010
/mod7/re...	0101	0101
/mod7/S	-107374...	-1073741825
/mod7/pa...	1101111...	11011111111111111111111111111111
/mod7/re...	{000} {0...	{000} {000} {000} {000} {000} {000} {000} {000} {000} {000} {000} {101}
/mod7/su...	0000101	0000101
/mod7/su...	{0000000...	{0000000} {0000000} {0000000} {0000000} {0000000} {0000101}
/mod7/su...	{0000000...	{0000000} {0000000} {0000101}
/mod7/su...	0000101	0000101
/mod7/su...	0000000	0000000
/mod7/ne...	1	

Τώρα χρειάζεται απλά με βάση τον πίνακα mod2 με mod7 να υπολογιστεί το mod14

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity mod14 is
    generic (n: integer := 32);
    port(
        A, B: in std_logic_vector(n-1 downto 0);
        remainder: out std_logic_vector(3 downto 0) -- [4-0] signed
    );
end mod14;

architecture dataflow of mod14 is

    signal S: std_logic_vector(n-1 downto 0);
    signal mod7_remainder: std_logic_vector(3 downto 0);

    component mod7 is
        generic (n: integer := 32);
        port(
            A, B: in std_logic_vector(n-1 downto 0);
            remainder: out std_logic_vector(3 downto 0) -- [3-0] signed
        );
    end component;

    component adder_n is
        generic (n: integer := 4);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
        );
    end component;

begin

    adder : adder_n
        generic map(n)
        port map (A, B, '0', S, open);
    mod7_module: mod7
        generic map(n)
        port map (A, B, mod7_remainder);
```

```

remainder <= "0000" when (S(0) = '0' and unsigned(mod7_remainder) = 0)
else -- 0
      "1000" when (S(0) = '0' and unsigned(mod7_remainder) = 1)
else -- 8
      "0010" when (S(0) = '0' and unsigned(mod7_remainder) = 2)
else -- 2
      "1010" when (S(0) = '0' and unsigned(mod7_remainder) = 3)
else -- 10
      "0100" when (S(0) = '0' and unsigned(mod7_remainder) = 4)
else -- 4
      "1100" when (S(0) = '0' and unsigned(mod7_remainder) = 5)
else -- 12
      "0110" when (S(0) = '0' and unsigned(mod7_remainder) = 6)
else -- 6

      "0111" when (S(0) = '1' and unsigned(mod7_remainder) = 0)
else -- 7
      "0001" when (S(0) = '1' and unsigned(mod7_remainder) = 1)
else -- 1
      "1001" when (S(0) = '1' and unsigned(mod7_remainder) = 2)
else -- 9
      "0011" when (S(0) = '1' and unsigned(mod7_remainder) = 3)
else -- 3
      "1011" when (S(0) = '1' and unsigned(mod7_remainder) = 4)
else -- 11
      "0101" when (S(0) = '1' and unsigned(mod7_remainder) = 5)
else -- 5
      "1101" when (S(0) = '1' and unsigned(mod7_remainder) = 6)
else -- 13

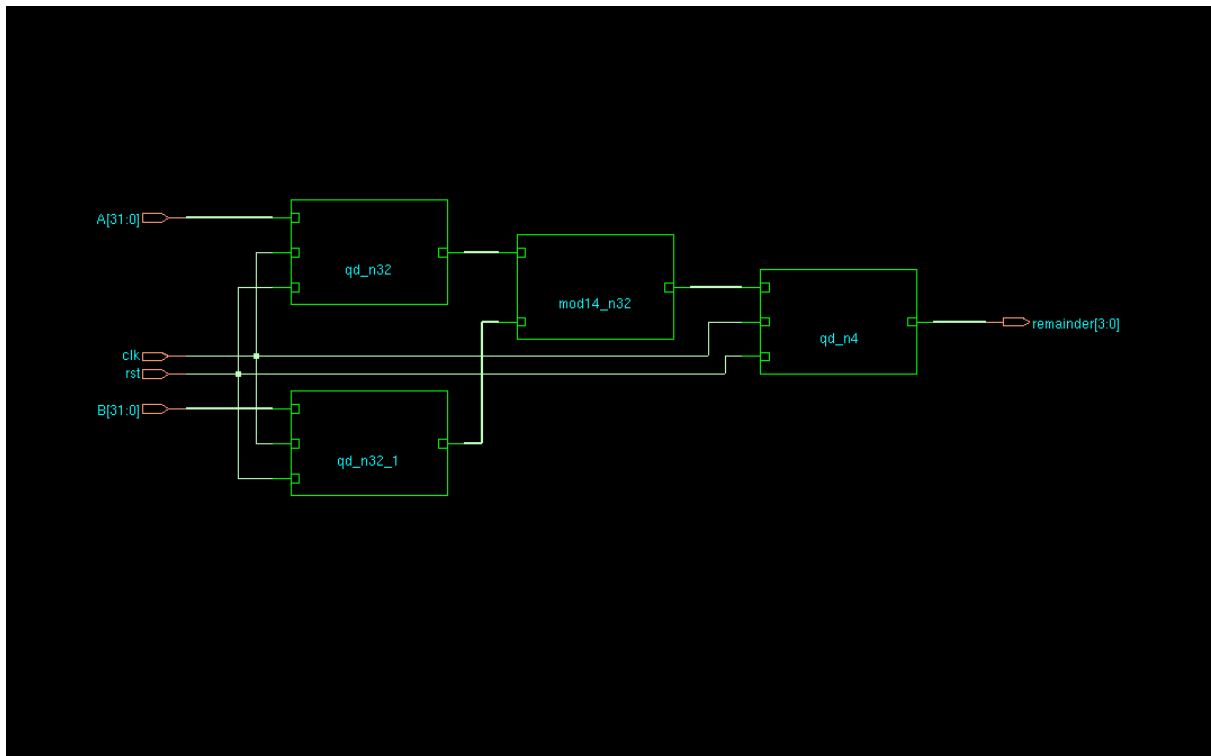
      "1111"; -- default fallback (error or undefined)

end dataflow;

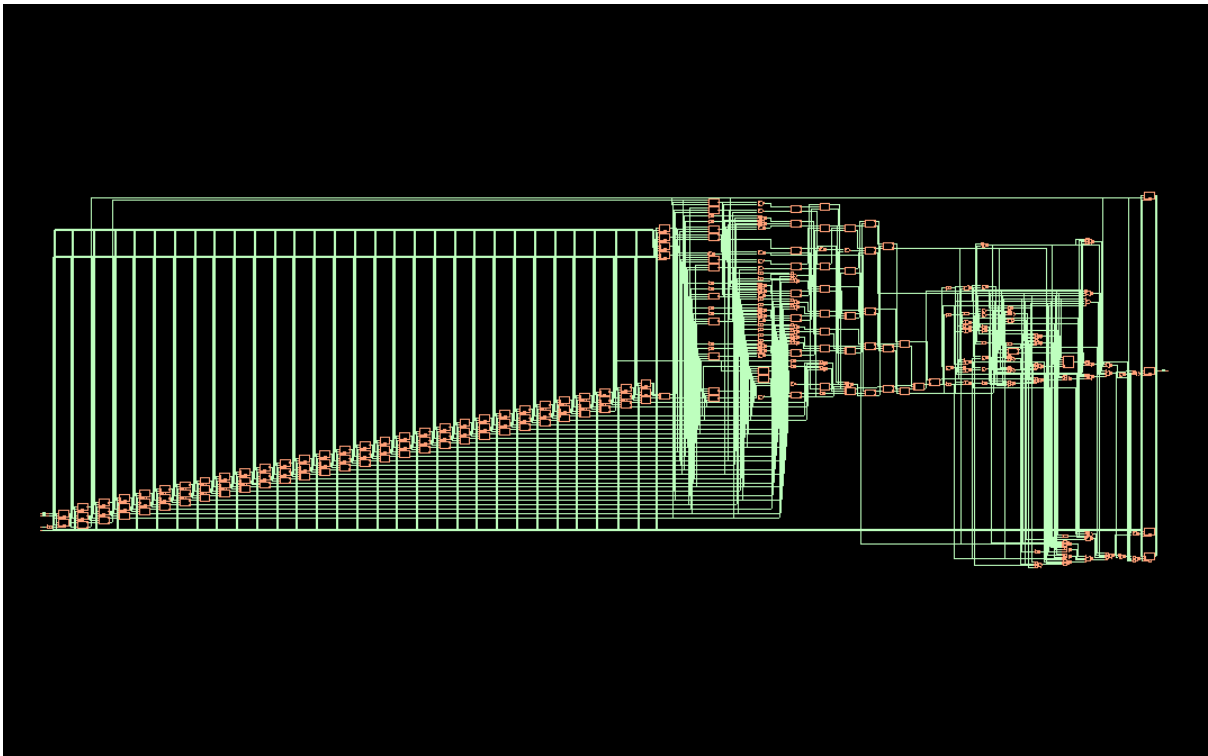
```

◆ /testbenc...	-715827883	216		-92		715827882	-715827883	2147483647	
◆ /testbenc...	-357913941	300		100		357913941	-357913941	-1	
◆ /testbenc...	1								
◆ /testbenc...	0								
◆ /testbenc...	7	X	15	12		8	7	6	0

Genus Modulo 14 Adder $\mu\epsilon$ Flip Flops



Genus Modulo 14 Adder $\mu\epsilon$ Flip Flops 45nm

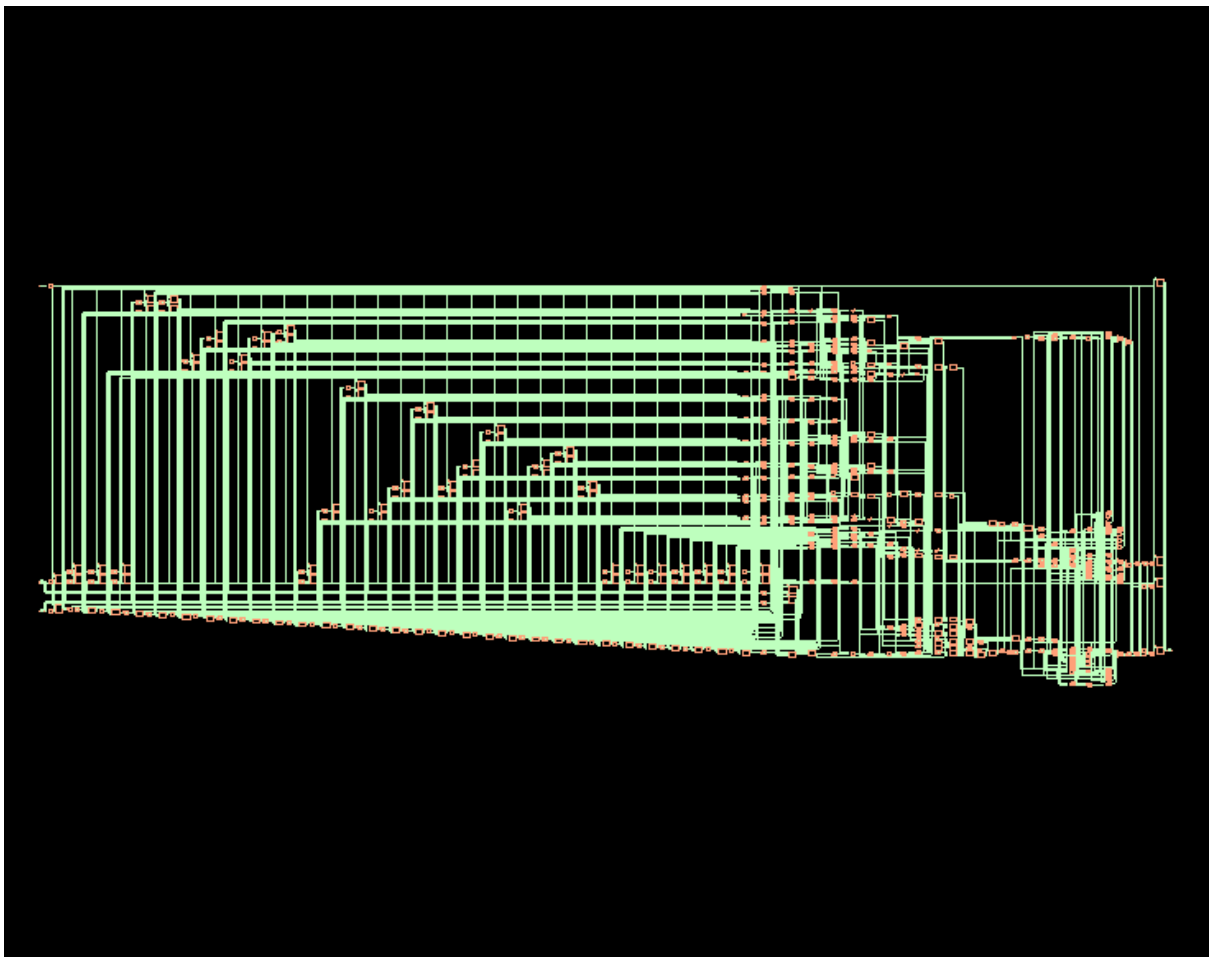


Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.98685e-08	5.67344e-05	4.61628e-06	6.13706e-05	48.01%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	2.21267e-08	2.55002e-05	3.85710e-05	6.40933e-05	50.14%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	2.36057e-06	2.36057e-06	1.85%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	4.19952e-08	8.22346e-05	4.55478e-05	1.27824e-04	100.00%
Percentage	0.03%	64.33%	35.63%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
#											
	b_qd_d_reg[0]/CK	-	-	R	(arrival)	68	-	0	0	0	(-, -)
	b_qd_d_reg[0]/Q	-	CK->Q	R	DFFRHQX1	1	5.2	101	292	292	(-, -)
	g6000_1666/C0	-	A->C0	R	ADDHX1	1	5.8	111	182	473	(-, -)
	g5997_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	247	720	(-, -)
	g5995_2883/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	973	(-, -)
	g5991_4733/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1225	(-, -)
	g5990_7482/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1478	(-, -)
	g5988_5115/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1730	(-, -)
	g5982_1705/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1983	(-, -)
	g5980_1617/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2235	(-, -)
	g5977_6783/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2488	(-, -)
	g5971_6260/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2740	(-, -)
	g5970_5107/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2993	(-, -)
	g5968_2398/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3245	(-, -)
	g5962_2883/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3498	(-, -)
	g5960_6161/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3750	(-, -)
	g5956_5115/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4003	(-, -)
	g5950_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4255	(-, -)
	g5948_2802/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4508	(-, -)
	g5945_1617/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4760	(-, -)
	g5938_6260/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5013	(-, -)
	g5936_2398/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5265	(-, -)
	g5933_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5518	(-, -)
	g5927_2883/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5770	(-, -)
	g5926_9945/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6023	(-, -)
	g5924_9315/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6275	(-, -)
	g5918_6131/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6528	(-, -)
	g5916_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6780	(-, -)
	g5913_3680/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7033	(-, -)
	g5906_5107/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7285	(-, -)
	g5904_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7538	(-, -)
	g5902_1666/S	-	CI->S	F	ADDFX1	3	9.6	196	380	7918	(-, -)
	g5897_9945/Y	-	A1N->Y	F	OAI2BB1X1	2	8.2	293	263	8181	(-, -)
	g5890_5115/S	-	B->S	R	ADDHX1	1	5.1	100	244	8425	(-, -)
	g5889_6131/Y	-	B->Y	F	CLKX0R2X1	2	7.7	154	237	8662	(-, -)
	g5885_5122/Y	-	B->Y	R	XNOR2X1	1	5.1	104	240	8902	(-, -)
	g5881_1617/Y	-	B->Y	F	CLKX0R2X1	1	5.8	124	222	9124	(-, -)
	g5876_8428/C0	-	CI->C0	F	ADDFX1	1	5.8	141	244	9368	(-, -)
	g5881_1617/Y	-	B->Y	F	CLKX0R2X1	1	5.8	124	222	9124	(-, -)
	g5876_8428/C0	-	CI->C0	F	ADDFX1	1	5.8	141	244	9368	(-, -)
	g5873_6260/C0	-	CI->C0	F	ADDFX1	1	5.8	142	250	9619	(-, -)
	g5865_7410/C0	-	CI->C0	F	ADDFX1	1	5.8	142	250	9869	(-, -)
	g5855_6161/S	-	CI->S	R	ADDFX1	5	15.8	266	428	10298	(-, -)
	g5854/Y	-	A->Y	F	INVX1	10	27.5	468	333	10631	(-, -)
	g5841_8246/Y	-	A->Y	F	AND2X1	4	12.7	223	308	10939	(-, -)
	g5838/Y	-	A->Y	R	INVX1	1	4.7	103	123	11062	(-, -)
	g5828_1666/Y	-	A1->Y	F	OAI22X1	1	4.8	219	168	11230	(-, -)
	g5816_6131/Y	-	B1->Y	R	A0I221X1	1	4.8	302	245	11476	(-, -)
	g5814_8246/Y	-	B0->Y	F	OAI2BB1X1	1	4.8	196	208	11683	(-, -)
	g5811_1617/Y	-	C0->Y	R	A0I221X1	1	4.5	289	207	11891	(-, -)
	g5809_3680/Y	-	B0->Y	R	0A21X1	2	7.4	131	257	12147	(-, -)
	g5807_8428/Y	-	B->Y	F	NAND2X1	3	9.9	348	226	12374	(-, -)
	g5806_4319/Y	-	B->Y	R	NOR2X1	2	7.4	269	239	12613	(-, -)
	g5801_2398/Y	-	B->Y	R	XNOR2X1	1	4.7	98	226	12839	(-, -)
	rem_qd_d_reg[2]/D <<<	-	-	R	DFFRHQX1	1	-	-	0	12839	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area
mod_flip_flop		267	939.816	358.128	1297.944

Genus Modulo 14 Adder μ e Flip Flops 7nm



Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	2.07323e-08	2.12190e-06	4.84365e-07	2.62700e-06	20.30%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.84649e-08	4.30525e-06	5.95011e-06	1.03138e-05	79.70%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	7.91972e-08	6.42716e-06	6.43448e-06	1.29408e-05	100.00%
Percentage	0.61%	49.67%	49.72%	100.00%	100.00%

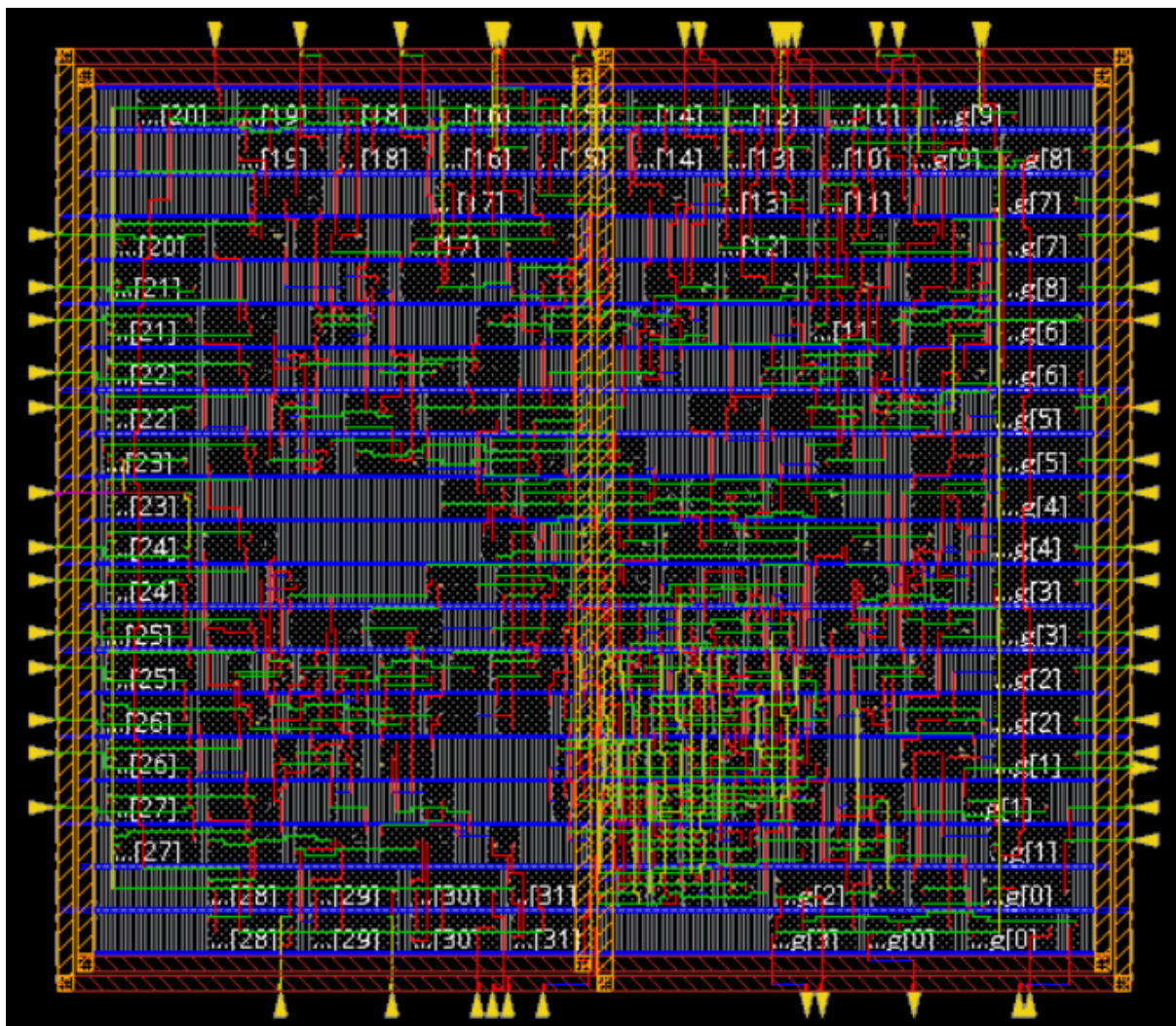
#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#	b_qd_d_reg[0]/CLK	-	-	R	(arrival)	68	-	0	0	0	(-, -)
#	b_qd_d_reg[0]/QN	-	CLK->QN	F	DFFASRHQNx1_ASAP7_75t_R	2	0.6	20	50	50	(-, -)
#	g6592_8428/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	2	1.3	42	28	78	(-, -)
#	g6591/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	2	1.1	41	35	113	(-, -)
#	g6558_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	38	30	143	(-, -)
#	g6557/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	28	26	169	(-, -)
#	g6554_2346/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.8	43	30	199	(-, -)
#	g6552/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	29	28	227	(-, -)
#	g6547_5115/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	36	26	253	(-, -)
#	g6545/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	26	279	(-, -)
#	g6540_6131/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	304	(-, -)
#	g6539/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	329	(-, -)
#	g6536_5122/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	355	(-, -)
#	g6535/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	380	(-, -)
#	g6530_8428/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	405	(-, -)
#	g6528/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	430	(-, -)
#	g6523_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	456	(-, -)
#	g6521/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	481	(-, -)
#	g6514_9315/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	507	(-, -)
#	g6513/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	532	(-, -)
#	g6506_8246/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	557	(-, -)
#	g6504/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	582	(-, -)
#	g6501_1705/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	608	(-, -)
#	g6500/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	633	(-, -)
#	g6497_3680/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	658	(-, -)
#	g6496/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	684	(-, -)
#	g6491_2398/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	709	(-, -)
#	g6489/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	734	(-, -)
#	g6484_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	760	(-, -)
#	g6482/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	785	(-, -)
#	g6475_6131/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	810	(-, -)
#	g6473/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	835	(-, -)
#	g6462_6260/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	861	(-, -)
#	g6460/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	886	(-, -)
#	g6453_2346/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	911	(-, -)
#	g6452/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	936	(-, -)

g6452/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	936	(-, -)
g6445_9315/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	962	(-, -)
g6444/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	987	(-, -)
g6438_7098/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1013	(-, -)
g6436/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1038	(-, -)
g6431_3680/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1063	(-, -)
g6429/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1088	(-, -)
g6423_5107/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1114	(-, -)
g6421/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1139	(-, -)
g6413_9945/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1164	(-, -)
g6411/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1190	(-, -)
g6408_6161/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1215	(-, -)
g6407/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1240	(-, -)
g6404_5115/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1266	(-, -)
g6403/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1291	(-, -)
g6398_2802/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1316	(-, -)
g6396/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1341	(-, -)
g6391_4319/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1367	(-, -)
g6389/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1392	(-, -)
g6382_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1417	(-, -)
g6380/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1442	(-, -)
g6370_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1468	(-, -)
g6367/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1493	(-, -)
g6360_2398/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1519	(-, -)
g6359/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1544	(-, -)
g6352_6161/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.3	35	26	1569	(-, -)
g6351/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.6	27	25	1594	(-, -)
g6340_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	1	1.5	38	27	1622	(-, -)
g6801/SN	-	CI->SN	F	FAX1_ASAP7_75t_R	3	2.9	65	88	1710	(-, -)
g6326/Y	-	A->Y	R	INVxp33_ASAP7_75t_R	2	0.8	52	42	1752	(-, -)
g6322_5477/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	3	2.1	52	42	1794	(-, -)
g6317_7410/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	1.3	46	36	1830	(-, -)
g6316/Y	-	A->Y	F	INVxp33_ASAP7_75t_R	1	0.3	22	22	1852	(-, -)
g6313_9945/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	2.1	59	36	1888	(-, -)
g6310_6161/Y	-	B->Y	F	NOR2xp33_ASAP7_75t_R	1	0.3	24	20	1908	(-, -)
g6307_1881/Y	-	B->Y	R	NOR2xp33_ASAP7_75t_R	2	2.4	71	40	1948	(-, -)
g6797/SN	-	A->SN	F	FAX1_ASAP7_75t_R	2	2.8	74	95	2042	(-, -)
g6295_6783/Y	-	C->Y	F	MAJx2_ASAP7_75t_R	2	2.4	17	46	2088	(-, -)
g6287_6260/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	1	1.5	14	30	2118	(-, -)
g6811/CON	-	CI->CON	R	FAX1_ASAP7_75t_R	2	0.8	38	22	2140	(-, -)

g6811/CON	-	C1->CON	R	FAX1 ASAP7_75t_R	2	0.8	38	22	2140	(-, -)
g2/Y	-	B->Y	R	AND2x2 ASAP7_75t_R	5	1.9	14	30	2170	(-, -)
g6260_6783/Y	-	B->Y	F	NOR2xp33 ASAP7_75t_R	4	1.7	40	26	2195	(-, -)
g6257/Y	-	A->Y	R	INVxp33 ASAP7_75t_R	8	3.7	162	90	2286	(-, -)
g6251_1666/Y	-	B->Y	F	NOR2xp33 ASAP7_75t_R	4	1.6	68	61	2346	(-, -)
g6247/Y	-	A->Y	R	INVxp33 ASAP7_75t_R	2	0.6	46	38	2384	(-, -)
g6242_5526/Y	-	B->Y	F	NOR2xp33 ASAP7_75t_R	1	0.3	20	18	2402	(-, -)
g6224_1666/Y	-	B->Y	R	NOR2xp33 ASAP7_75t_R	1	0.3	17	14	2416	(-, -)
g6216_7482/Y	-	B->Y	F	NOR2xp33 ASAP7_75t_R	1	0.5	33	14	2430	(-, -)
g6208_5122/Y	-	A->Y	R	NOR4xp25 ASAP7_75t_R	1	0.5	40	26	2457	(-, -)
g6206_1705/Y	-	A->Y	F	NAND5xp2 ASAP7_75t_R	2	1.3	58	37	2494	(-, -)
g6205/Y	-	A->Y	R	INVxp33 ASAP7_75t_R	1	0.3	32	28	2521	(-, -)
g6202_6783/Y	-	B->Y	F	NAND2xp33 ASAP7_75t_R	1	0.5	22	16	2537	(-, -)
g6198_4319/Y	-	A->Y	F	AND2x2 ASAP7_75t_R	1	0.3	7	24	2561	(-, -)
g6196_2398/Y	-	B->Y	R	NOR2xp33 ASAP7_75t_R	1	0.6	21	14	2574	(-, -)
rem_qd_d_reg[3]/D	-	-	R	DFFASRH0Nx1 ASAP7_75t_R	1	-	-	0	2574	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
mod_flip_flop		507	62.475	0.000	62.475	<none> (D)

Innovus για Modulo 14 adder με Flip Flops 45nm



```
# Generated on: Tue Jul 8 15:27:26 2025
# Design: mod_flip_flop
# Command: report_timing
#####
Path 1: MET Setup Check with Pin rem_qd_d_reg[2]/CK
Endpoint: rem_qd_d_reg[2]/D (^) checked with leading edge of 'clk'
Beginpoint: b_qd_d_reg[0]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: worst_case
Other End Arrival Time -0.000
- Setup 0.108
+ Phase Shift 14.000
= Required Time 13.892
- Arrival Time 9.688
= Slack Time 4.205
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.000
= Beginpoint Arrival Time -0.000
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
b_qd_d_reg[0]	CK ^			-0.000	4.204
b_qd_d_reg[0]	CK ^ -> Q ^	DFFRHQX1	0.269	0.269	4.474
g6000_1666	A ^ -> C0 ^	ADDFX1	0.141	0.410	4.615
g5997_2346	CI ^ -> C0 ^	ADDFX1	0.204	0.614	4.819
g5995_2883	CI ^ -> C0 ^	ADDFX1	0.209	0.823	5.028
g5991_4733	CI ^ -> C0 ^	ADDFX1	0.215	1.039	5.243
g5990_7482	CI ^ -> C0 ^	ADDFX1	0.212	1.250	5.455
g5988_5115	CI ^ -> C0 ^	ADDFX1	0.209	1.459	5.664
g5982_1705	CI ^ -> C0 ^	ADDFX1	0.209	1.668	5.872
g5980_1617	CI ^ -> C0 ^	ADDFX1	0.208	1.876	6.080
g5977_6783	CI ^ -> C0 ^	ADDFX1	0.208	2.084	6.288
g5971_6260	CI ^ -> C0 ^	ADDFX1	0.209	2.293	6.497
g5970_5107	CI ^ -> C0 ^	ADDFX1	0.210	2.503	6.707
g5968_2398	CI ^ -> C0 ^	ADDFX1	0.209	2.712	6.917
g5962_2883	CI ^ -> C0 ^	ADDFX1	0.208	2.920	7.125
g5960_6161	CI ^ -> C0 ^	ADDFX1	0.208	3.129	7.333
g5956_5115	CI ^ -> C0 ^	ADDFX1	0.208	3.337	7.541
g5950_5122	CI ^ -> C0 ^	ADDFX1	0.207	3.544	7.749
g5948_2802	CI ^ -> C0 ^	ADDFX1	0.209	3.753	7.958
g5945_1617	CI ^ -> C0 ^	ADDFX1	0.210	3.964	8.168

g5948_2802	CI ^ -> C0 ^	ADDFX1	0.209	3.753	7.958
g5945_1617	CI ^ -> C0 ^	ADDFX1	0.210	3.964	8.168
g5938_6260	CI ^ -> C0 ^	ADDFX1	0.208	4.172	8.376
g5936_2398	CI ^ -> C0 ^	ADDFX1	0.207	4.379	8.584
g5933_6417	CI ^ -> C0 ^	ADDFX1	0.209	4.588	8.793
g5927_2883	CI ^ -> C0 ^	ADDFX1	0.209	4.797	9.002
g5926_9945	CI ^ -> C0 ^	ADDFX1	0.210	5.008	9.212
g5924_9315	CI ^ -> C0 ^	ADDFX1	0.210	5.218	9.423
g5918_6131	CI ^ -> C0 ^	ADDFX1	0.210	5.428	9.632
g5916_5122	CI ^ -> C0 ^	ADDFX1	0.210	5.638	9.843
g5913_3680	CI ^ -> C0 ^	ADDFX1	0.212	5.850	10.054
g5906_5107	CI ^ -> C0 ^	ADDFX1	0.210	6.060	10.264
g5904_6417	CI ^ -> C0 ^	ADDFX1	0.211	6.271	10.475
g5902_1666	CI ^ -> S v	ADDFX1	0.308	6.579	10.783
g5897_9945	A1N v -> Y v	OAI2BB1X1	0.143	6.722	10.926
g5890_5115	B v -> S ^	ADDFX1	0.166	6.888	11.093
g5889_6131	B ^ -> Y v	CLKX0R2X1	0.176	7.064	11.269
g5885_5122	B v -> Y ^	XNOR2X1	0.188	7.253	11.457
g5881_1617	B ^ -> Y v	CLKX0R2X1	0.177	7.429	11.634
g5876_8428	CI v -> C0 v	ADDFX1	0.198	7.627	11.832
g5873_6260	CI v -> C0 v	ADDFX1	0.204	7.831	12.035
g5865_7410	CI v -> C0 v	ADDFX1	0.206	8.037	12.241
g5855_6161	CI v -> S ^	ADDFX1	0.354	8.390	12.595
g5854	A ^ -> Y v	INVX1	0.151	8.541	12.745
g5842_1705	A v -> Y v	AND2X1	0.150	8.691	12.895
g5826_2346	AN v -> Y v	NOR3BX1	0.088	8.779	12.984
g5821_4733	B v -> Y v	OR2X1	0.102	8.881	13.086
g5816_6131	A1 v -> Y ^	A0I221X1	0.121	9.002	13.207
g5814_8246	B0 ^ -> Y v	OAI2BB1X1	0.099	9.102	13.306
g5811_1617	C0 v -> Y ^	A0I221X1	0.093	9.194	13.399
g5809_3680	B0 ^ -> Y ^	OA21X1	0.167	9.362	13.566
g5807_8428	B ^ -> Y v	NAND2X1	0.093	9.455	13.659
g5806_4319	B v -> Y ^	NOR2X1	0.094	9.549	13.753
g5801_2398	B ^ -> Y ^	XNOR2X1	0.139	9.688	13.892
rem_qd_d_reg[2]	D ^	DFFRHQX1	0.000	9.688	13.892

LEC για Modulo 14 adder με Flip Flops

Verification Report	
Category	Count
1. Non-standard modeling options used:	0
2. Incomplete verification:	0
3. User modification to design:	0
4. Conformal Constraint Designer clock domain crossing checks recommended:	0
5. Design ambiguity:	0
6. Compare Results:	PASS

Xcelium για Modulo 14 adder με Flip Flops



Modulo 15 Adder

Συγκριτής

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity comparator_of_15 is
  generic (n: integer := 7);
  port (
    S: in std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
  );
end comparator_of_15;

architecture dataflow of comparator_of_15 is
  signal test: std_logic_vector(3 downto 0); -- 4 bits (but we only
compare 3)
  signal above_below: std_logic_vector(1 downto 0);
  signal S_padded : std_logic_vector(6 downto 0); -- always 7 bits
begin
  S_padded <= (6 downto n => '0') & S;
  test <= std_logic_vector(to_unsigned(0, 4)); -- "0000"

  above_below <= "10" when (
    (S_padded(6 downto 4) /= "000" and negative = '0')
or
    (negative = '0' and S(3) = '1' and S(2) = '1' and
S(1) = '1' and S(0) = '1')
  ) else
    "01" when (S(n-1) = '1' and negative = '1') else
    "00";

  above <= above_below(1);
  below <= above_below(0);
end dataflow;
```

/comparat...	7	7						
+ /comparat...	0001111	1000000		0001000		0001111		
/comparat...	0							
/comparat...	1							
/comparat...	0							
+ /comparat...	0000	0000						
+ /comparat...	10	10		00		10		
+ /comparat...	0001111	1000000		0001000		0001111		

mod15 chunking

```

library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity remainder_calculator_15 is generic (n:integer:=4);
  port (
    data_in      : in  std_logic_vector(n-1 downto 0);
    negative_first: in  std_logic;
    remainder     : out std_logic_vector(n-1 downto 0)
  );
end entity;

architecture rtl of remainder_calculator_15 is

  signal mod15, mod15_comp : std_logic_vector(6 downto 0);
  constant all_ones : std_logic_vector(n-1 downto 0) := (others =>
'1');

  type slv6_array is array (natural range <>) of std_logic_vector(6
downto 0);
  signal data : slv6_array(0 to 8);

  component comparator_of_15 is generic (n:integer:=4); port(
    S: in  std_logic_vector(n-1 downto 0);
    negative: in std_logic;
    above: out std_logic;
    below: out std_logic
  );
end component;

```

```

component adder_n is
    generic (n: integer := 4);
    port(
        a, b: in std_logic_vector(n-1 downto 0);
        cin: in std_logic;
        sum: out std_logic_vector(n-1 downto 0);
        cout: out std_logic
    );
end component;

begin
    mod15 <= std_logic_vector(to_signed(15, 7));
    mod15_comp <= std_logic_vector(to_signed(-15, 7));

    data(0) <= (others => '0') when (data_in = all_ones and negative_first =
    '0') else
        (6 downto n => data_in(n-1)) & data_in when negative_first
    ='1' else
        (6 downto n => '0') & data_in;

    -- so that for 15 when mod15_comp can't be actually used for 4 bits so i
    bypass it
    -- for 7 bits 1111111 in the adding remainders stage, its impossible as
    max 14*8=112=1110000
    generate_label:
    for i in 0 to 7 generate
        signal above_i, below_i: std_logic;
        signal temp1_i, temp2_i: std_logic_vector(6 downto 0);
    begin
        -- First comparison stage
        comp_i: comparator_of_15
            generic map(7)
            port map(data(i), negative_first, above_i, below_i);

        -- First arithmetic stage
        add_above_i: adder_n
            generic map(7)
            port map(data(i), mod15_comp, '0', temp1_i, open);

        add_below_i: adder_n
            generic map(7)
            port map(data(i), mod15, '0', temp2_i, open);

        -- Intermediate mux

```

```

data(i+1) <= temp1_i when above_i = '1' else
               temp2_i when below_i = '1' else
               data(i);

end generate;

-- Final output selection
remainder<=data(8)(n-1 downto 0);

end architecture;

```

/remainder_calculat...	4	4																		
/remainder_calculat...	1111	1111																		
/remainder_calculat...	0																			
/remainder_calculat...	0000	1110						0000												
/remainder_calculat...	0001111	0001111																		
/remainder_calculat...	1110001	1110001																		
/remainder_calculat...	{0000000...}	{1111111} {0001110} {0001...}						{0000000} {0000000} {0000...												
/remainder_calculat...	1111	1111																		

/remainder_calculator_15/n	7	7																		
/remainder_calculator_15/data_in	1110000	1110000																		
/remainder_calculator_15/negative_first	0																			
/remainder_calculator_15/remainder	0000111	0000111																		
/remainder_calculator_15/mod15	0001111	0001111																		
/remainder_calculator_15/mod15_comp	1110001	1110001																		
/remainder_calculator_15/data	{-16} {-3...}	{-16} {-31} {-46} {-61} {52} {37} {22} {7} {7}																		
/remainder_calculator_15/all_ones	1111111	1111111																		

mod15 final calculation

```

library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity mod15 is
    generic (n: integer := 32);
    port(
        A, B: in std_logic_vector(n-1 downto 0);
        remainder: out std_logic_vector(3 downto 0) -- [4-0] signed
    );
end entity;

```

```

end mod15;

architecture dataflow of mod15 is

    signal S: std_logic_vector(n-1 downto 0);
    signal padded_32 : std_logic_vector(31 downto 0);

    -- Correct array type declaration
    type slv4_array is array (natural range <>) of std_logic_vector(3
downto 0);
    signal remainders : slv4_array(0 to 7);
    signal sum_remain : std_logic_vector(6 downto 0);
    type slv7_array is array (natural range <>) of std_logic_vector(6
downto 0);
    -- Adder tree signals
    signal sum_stage1 : slv7_array(0 to 3);
    signal sum_stage2 : slv7_array(0 to 1);
    signal sum_final : std_logic_vector(6 downto 0);

    -- decide if i want to calculate as if its a negative or positive
number
    signal negative:std_logic;
    signal s_first:std_logic_vector(3 downto 0);
    -- Component declarations
    component adder_n is
        generic (n: integer := 4);
        port(
            a, b: in std_logic_vector(n-1 downto 0);
            cin: in std_logic;
            sum: out std_logic_vector(n-1 downto 0);
            cout: out std_logic
        );
    end component;

    component comparator_of_15 is
        generic (n: integer := 4);
        port(
            S: in std_logic_vector(n-1 downto 0);
            negative: in std_logic;
            above: out std_logic;
            below: out std_logic
        );
    end component;

    component remainder_calculator_15 is generic (n: integer := 4);
    port (

```

```

        data_in      : in  std_logic_vector(n-1 downto 0);
        negative_first: in  std_logic;
        remainder     : out std_logic_vector(n-1 downto 0)
    );
end component;

begin

adder : adder_n
    generic map(n)
    port map (A, B, '0', S, open);

padded_32 <= (31 downto n => S(n-1)) & S;

negative <= padded_32(31);

s_first <= padded_32(31 downto 28);

remainder_c : remainder_calculator_15
    generic map(4)
    port map(s_first, negative,
        remainders(7));

generate_label:
for i in 0 to 6 generate

    signal s_in: std_logic_vector(3 downto 0);
    begin

s_in <= padded_32(3+i*4 downto i*4);

remainder_c : remainder_calculator_15
    generic map(4)
    port map(s_in, '0',
        remainders(i));
    end generate;

-- Adder tree to sum all remainders
-- Stage 1: Add pairs of remainders

gen_stage1: for i in 0 to 3 generate

```

```

    signal a_padded, b_padded : std_logic_vector(6 downto 0);
begin
    a_padded <= "000" & remainders(i*2);
    b_padded <= "000" & remainders(i*2+1);

    adder_st1: adder_n
        generic map(7)
        port map(
            a => a_padded,
            b => b_padded,
            cin => '0',
            sum => sum_stage1(i),
            cout => open
        );
end generate gen_stage1;

-- Stage 2: Add results from stage 1
gen_stage2: for i in 0 to 1 generate
    adder_st2: adder_n
        generic map(7)
        port map(
            a => sum_stage1(i*2),
            b => sum_stage1(i*2+1),
            cin => '0',
            sum => sum_stage2(i),
            cout => open
        );
end generate gen_stage2;

-- Final adder
adder_final: adder_n
    generic map(7)
    port map(
        a => sum_stage2(0),
        b => sum_stage2(1),
        cin => '0',
        sum => sum_final,
        cout => open
    );

remainder_sum : remainder_calculator_15
generic map(7)
port map(sum_final, '0',
sum_remain);

remainder <= sum_remain(3 downto 0);

```

```
end dataflow;
```

/mod15/n	32	32
/mod15/A	1861152494	1861152494
/mod15/B	0	0
/mod15/remainder	14	14
/mod15/S	1861152494	1861152494
/mod15/padded_32	0110111011...	01101110111011101110111011101110
/mod15/remainders	{1110} {11...	{1110} {1110} {1110} {1110} {1110} {1110} {1110} {0110}
/mod15/sum_remain	0001110	0001110
/mod15/sum_stage1	{0011100} ...	{0011100} {0011100} {0011100} {0010100}
/mod15/sum_stage2	{0111000} ...	{0111000} {0110000}
/mod15/sum_final	1101000	1101000
/mod15/negative	0	
/mod15/s_first	0110	0110

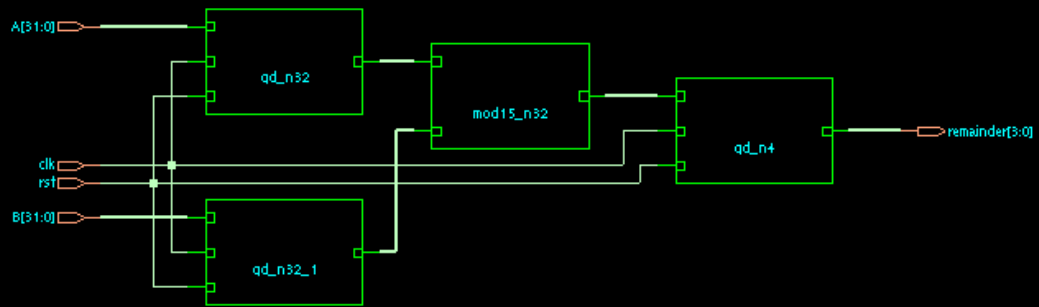
1 861 152 494 modulo 15 =

14

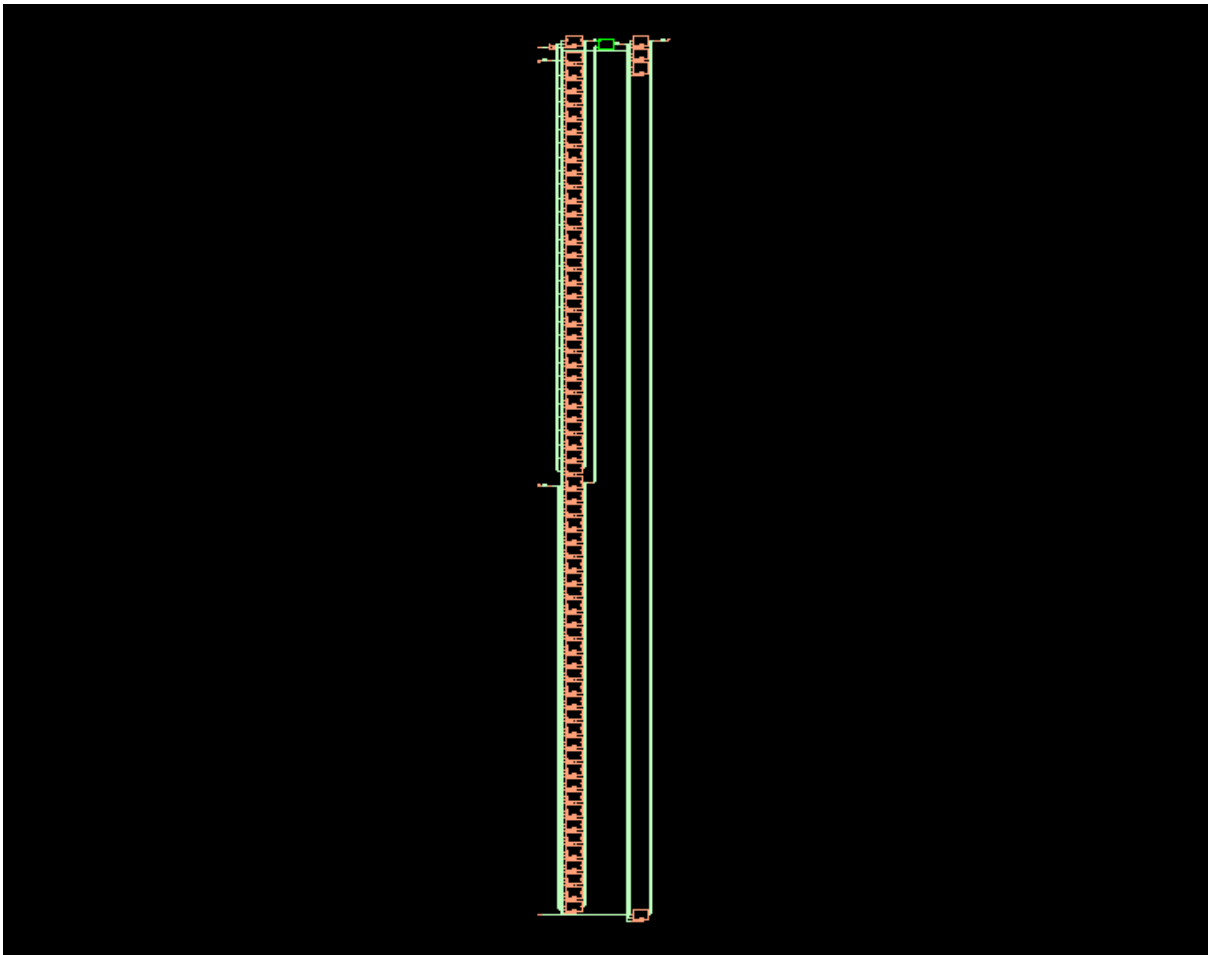
Τελικό testbench με flip flops

		Msgs									
[-] ◆ /testbenc...	715827882	216		-92		715827882		-715827883		2147483647	
[-] ◆ /testbenc...	357913941	300		100		357913941		-357913941		-1	
[-] ◆ /testbenc...	0										
[-] ◆ /testbenc...	0										
[+] ◆ /testbenc...	8	X	6		8		3		11		6

Genus Modulo 15 Adder $\mu\epsilon$ Flip Flops



Genus Modulo 15 Adder μ e Flip Flops 45nm

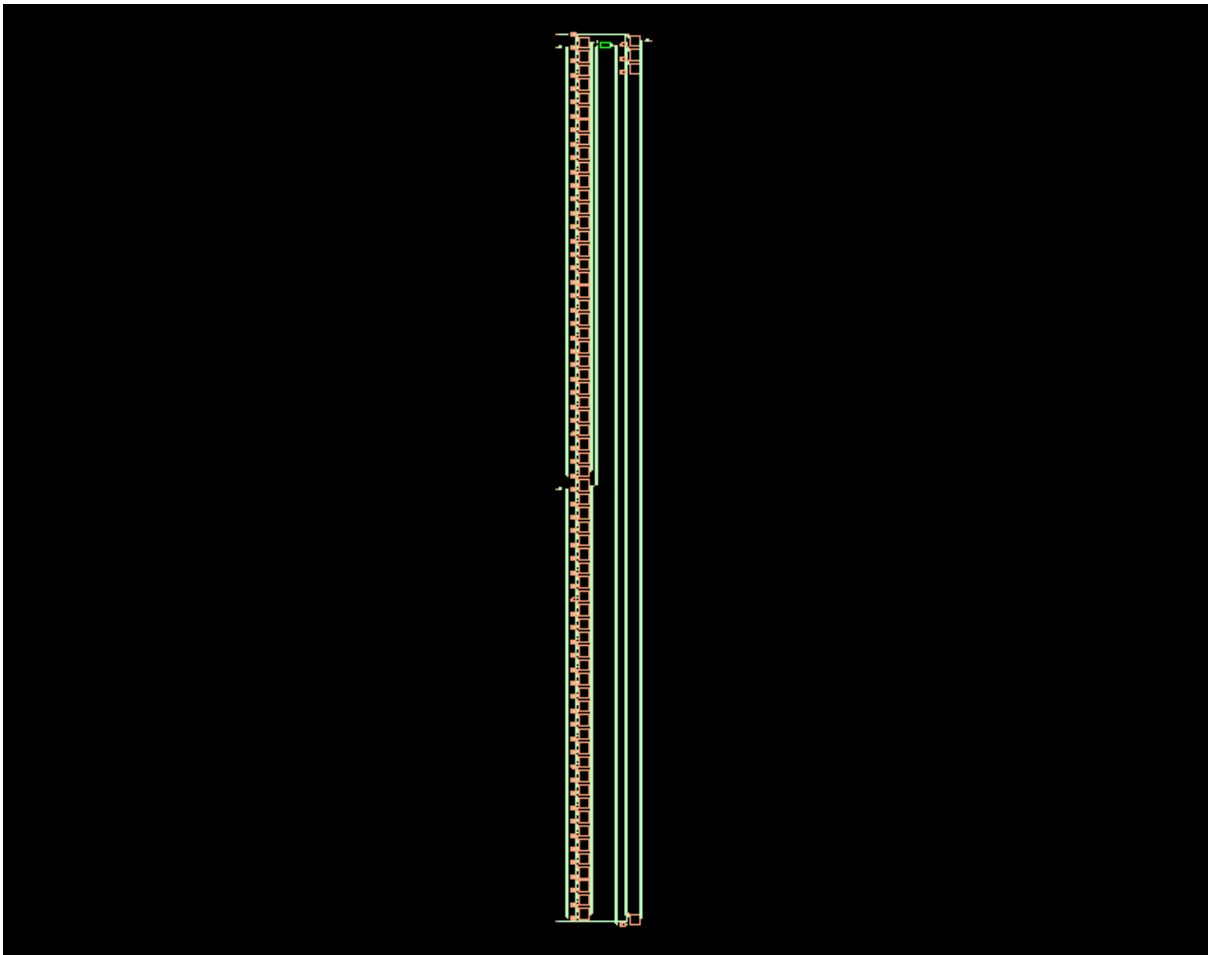


Category	Leakage	Internal	Switching	Total	Row%
memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	1.98685e-08	6.16110e-05	5.10923e-06	6.67401e-05	52.99%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	1.98114e-08	2.36394e-05	3.29957e-05	5.66550e-05	44.99%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	2.54215e-06	2.54215e-06	2.02%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
Subtotal	3.96799e-08	8.52505e-05	4.06471e-05	1.25937e-04	100.00%
Percentage	0.03%	67.69%	32.28%	100.00%	100.00%

#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
	b_qd_d_reg[0]/CK	-	-	R	(arrival)	68	-	0	0	0	(-, -)
	b_qd_d_reg[0]/Q	-	CK->Q	R	DFFRHQX1	1	5.2	101	292	292	(-, -)
	adder/g5485_2802/C0	-	A->C0	R	ADDFX1	1	5.8	111	182	473	(-, -)
	adder/g5484_1617/C0	-	CI->C0	R	ADDFX1	1	5.8	128	247	720	(-, -)
	adder/g5483_3680/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	973	(-, -)
	adder/g5481_5526/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1225	(-, -)
	adder/g5473_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1478	(-, -)
	adder/g5472_7410/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1730	(-, -)
	adder/g5471_1666/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	1983	(-, -)
	adder/g5469_2883/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2235	(-, -)
	adder/g5460_6131/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2488	(-, -)
	adder/g5458_8246/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2740	(-, -)
	adder/g5456_1705/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	2993	(-, -)
	adder/g5453_3680/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3245	(-, -)
	adder/g5444_5477/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3498	(-, -)
	adder/g5443_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	3750	(-, -)
	adder/g5442_7410/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4003	(-, -)
	adder/g5440_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4255	(-, -)
	adder/g5431_1881/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4508	(-, -)
	adder/g5429_7098/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	4760	(-, -)
	adder/g5426_1705/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5013	(-, -)
	adder/g5423_3680/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5265	(-, -)
	adder/g5413_6417/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5518	(-, -)
	adder/g5411_1666/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	5770	(-, -)
	adder/g5409_2883/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6023	(-, -)
	adder/g5406_6161/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6275	(-, -)
	adder/g5397_5122/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6528	(-, -)
	adder/g5395_2802/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	6780	(-, -)
	adder/g5393_3680/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7033	(-, -)
	adder/g5389_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7285	(-, -)
	adder/g5380_1666/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7538	(-, -)
	adder/g5379_2346/C0	-	CI->C0	R	ADDFX1	1	5.8	128	252	7790	(-, -)
	adder/g5378_2883/C0	-	CI->C0	R	ADDFX1	1	5.0	116	246	8037	(-, -)
	adder/g5377_9945/Y	-	B->Y	F	XNOR2X1	4	12.8	234	264	8301	(-, -)
	adder/g5375_6161/Y	-	B->Y	R	NAND2BX1	2	6.9	138	145	8446	(-, -)
	adder/g5372_5115/Y	-	B->Y	R	OR2X1	2	7.2	120	152	8598	(-, -)
	adder/g5371_1881/Y	-	B0->Y	F	OAI2BB1X1	1	5.8	218	164	8762	(-, -)
	adder/g5372_5115/Y	-	B->Y	R	OR2X1	2	7.2	120	152	8598	(-, -)
	adder/g5371_1881/Y	-	B0->Y	F	OAI2BB1X1	1	5.8	218	164	8762	(-, -)
	adder/g5367_5122/C0	-	CI->C0	F	ADDFX1	1	5.8	142	276	9038	(-, -)
	adder/g5365_2802/S	-	CI->S	R	ADDFX1	1	6.3	135	359	9396	(-, -)
	adder/g5362_6783/C0	-	B->C0	R	ADDFX1	1	5.8	127	271	9667	(-, -)
	adder/g5360_8428/C0	-	CI->C0	R	ADDFX1	1	5.8	127	252	9919	(-, -)
	adder/g5359_4319/S	-	CI->S	F	ADDFX1	1	6.3	145	352	10272	(-, -)
	adder/g5356_2398/S	-	B->S	R	ADDFX1	5	15.3	259	438	10710	(-, -)
	adder/g5352_6417/Y	-	B->Y	F	NAND2X1	5	15.1	508	353	11064	(-, -)
	adder/g5488/Y	-	A->Y	R	INVX1	2	7.5	193	251	11315	(-, -)
	adder/g5336_6131/Y	-	A0->Y	F	A0I21X1	2	7.2	280	225	11540	(-, -)
	adder/g5329_3680/Y	-	A1N->Y	F	A0I2BB1X1	1	4.5	110	215	11755	(-, -)
	adder/g5325_4319/Y	-	A->Y	R	MXI2XL	1	4.7	303	197	11952	(-, -)
	rem_qd_d_reg[3]/D	<<<	-	R	DFFRHQX1	1	-	-	0	11952	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area
mod_flip_flop		234	888.858	317.759	1206.617
adder	mod15 n32	165	469.566	216.952	686.518

Genus Modulo 15 Adder μ e Flip Flops 7nm



Category	Leakage	Internal	Switching	Total	Row%
memory	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
register	2.07327e-08	2.30756e-06	5.05680e-07	2.83397e-06	20.82%
latch	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
logic	8.18492e-08	4.80012e-06	5.89458e-06	1.07765e-05	79.18%
bbox	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
clock	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pad	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
pm	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.00%
Subtotal	1.02582e-07	7.10768e-06	6.40026e-06	1.36105e-05	100.00%
Percentage	0.75%	52.22%	47.02%	100.00%	100.00%

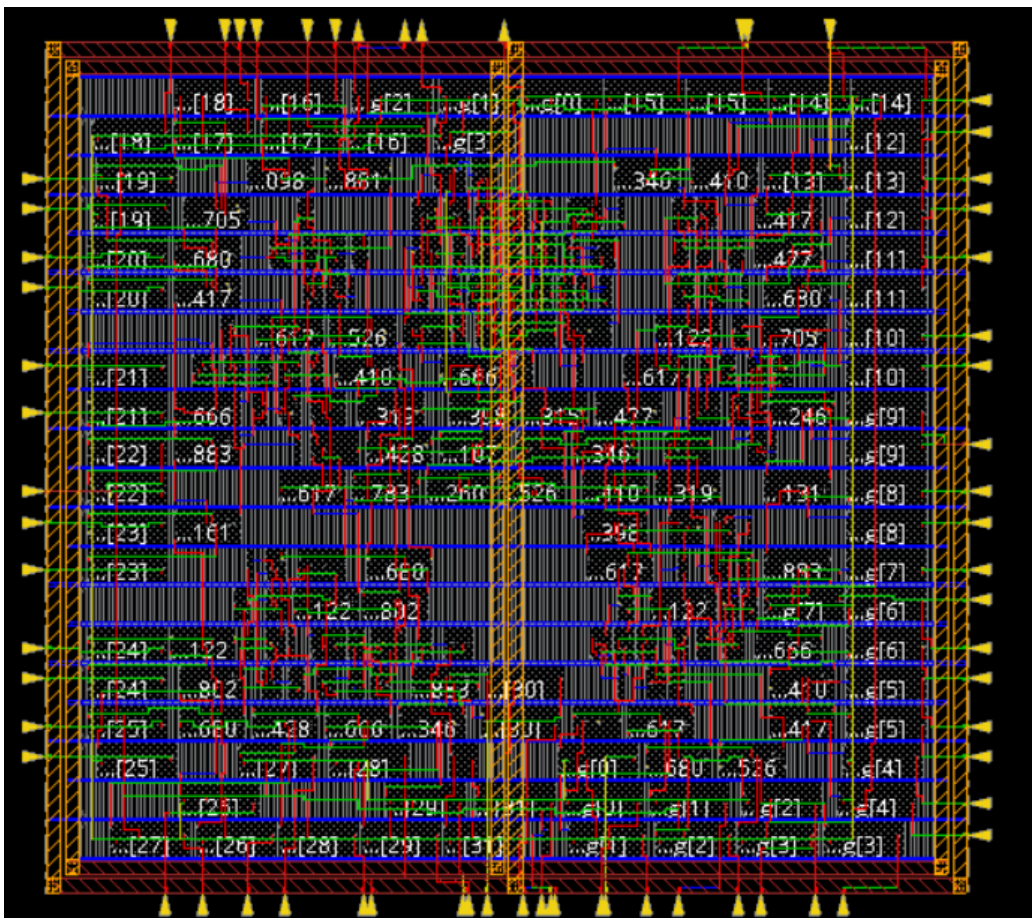
#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#											
#											
	b_qd_d_reg[0]/CLK	-	-	R	(arrival)	68	-	0	0	0	(-,-)
	b_qd_d_reg[0]/QN	-		CLK->QN F	DFFASRRHONx1_ASAP7_75t_R	2	0.6	20	50	50	(-,-)
	adder/g6134_2883/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	2	1.6	49	32	81	(-,-)
	adder/g6133/Y	-	A->Y	F	INVx1_ASAP7_75t_R	2	1.1	23	23	104	(-,-)
	adder/g6100_7482/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	31	135	(-,-)
	adder/g6099/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	153	(-,-)
	adder/g6097_1881/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	177	(-,-)
	adder/g6096/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	194	(-,-)
	adder/g6091_5122/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	218	(-,-)
	adder/g6090/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	235	(-,-)
	adder/g6085_6783/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	258	(-,-)
	adder/g6083/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	275	(-,-)
	adder/g6078_2398/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	304	(-,-)
	adder/g6077/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	322	(-,-)
	adder/g6075_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	346	(-,-)
	adder/g6074/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	363	(-,-)
	adder/g6069_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	387	(-,-)
	adder/g6068/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	404	(-,-)
	adder/g6061_1881/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	427	(-,-)
	adder/g6059/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	444	(-,-)
	adder/g6053_1617/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	468	(-,-)
	adder/g6051/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	484	(-,-)
	adder/g6046_6260/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	508	(-,-)
	adder/g6044/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	525	(-,-)
	adder/g6037_1666/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	549	(-,-)
	adder/g6036/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	565	(-,-)
	adder/g6027_5115/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	589	(-,-)
	adder/g6025/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	606	(-,-)
	adder/g6022_8246/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	629	(-,-)
	adder/g6021/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	646	(-,-)
	adder/g6019_1705/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	670	(-,-)
	adder/g6018/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	686	(-,-)
	adder/g6014_6783/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	710	(-,-)
	adder/g6013/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	727	(-,-)
	adder/g6006_5477/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	751	(-,-)
	adder/g6004/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	767	(-,-)
	adder/g5999_9945/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	796	(-,-)
	adder/g5996/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	814	(-,-)
	adder/g5990_6131/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	838	(-,-)
	adder/g5988/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	855	(-,-)
	adder/g5980_3680/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	879	(-,-)
	adder/g5977/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	896	(-,-)
	adder/g5964_9945/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	920	(-,-)
	adder/g5963/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	936	(-,-)
	adder/g5955_7098/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	965	(-,-)
	adder/g5954/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	983	(-,-)
	adder/g5950_2802/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	1007	(-,-)
	adder/g5949/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1024	(-,-)
	adder/g5943_8428/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	1053	(-,-)
	adder/g5942/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	1071	(-,-)
	adder/g5935_7410/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	1095	(-,-)
	adder/g5932/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1112	(-,-)
	adder/g5927_6161/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	1136	(-,-)
	adder/g5924/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	1153	(-,-)
	adder/g5919_8246/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	1181	(-,-)
	adder/g5918/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	1199	(-,-)
	adder/g5911_6783/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	39	24	1224	(-,-)
	adder/g5909/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1241	(-,-)
	adder/g5900_6417/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	1.5	38	24	1265	(-,-)
	adder/g5898/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	16	17	1281	(-,-)
	adder/g5894_2883/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	2	2.1	48	29	1310	(-,-)
	adder/g5893/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	18	18	1328	(-,-)
	adder/g5891_9315/Y	-	A->Y	R	MAJIXp5_ASAP7_75t_R	1	1.5	39	24	1352	(-,-)
	adder/g6142/SN	-	CI->SN	F	FAx1_ASAP7_75t_R	4	2.1	52	73	1426	(-,-)
	adder/g5885_5115/Y	-	B->Y	F	AND2x2_ASAP7_75t_R	2	0.8	9	32	1458	(-,-)
	adder/g5881_8246/Y	-	B->Y	F	OR2x2_ASAP7_75t_R	1	0.3	8	21	1479	(-,-)
	adder/g5879_1705/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	2	2.0	59	32	1511	(-,-)
	adder/g5873_5526/Y	-	B->Y	R	XOR2xp5_ASAP7_75t_R	1	1.0	33	35	1545	(-,-)
	adder/g5867_2398/Y	-	B->Y	R	XOR2xp5_ASAP7_75t_R	2	1.6	41	35	1580	(-,-)
	adder/g5866/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1598	(-,-)
	adder/g5859_2346/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	2	1.6	14	30	1628	(-,-)
	adder/g5854_4733/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	2	1.6	14	29	1656	(-,-)

adder/g5873_5526/Y	-	B->Y	R	XOR2xp5_ASAP7_75t_R	1	1.0	33	35	1545	(-, -)
adder/g5867_2398/Y	-	B->Y	R	XOR2xp5_ASAP7_75t_R	2	1.6	41	35	1580	(-, -)
adder/g5866/Y	-	A->Y	F	INVx1_ASAP7_75t_R	1	0.6	17	17	1598	(-, -)
adder/g5859_2346/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	2	1.6	14	30	1628	(-, -)
adder/g5854_4733/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	2	1.6	14	29	1656	(-, -)
adder/g5848_7098/Y	-	A->Y	F	MAJx2_ASAP7_75t_R	1	1.5	13	29	1685	(-, -)
adder/g6153/CON	-	CI->CON	R	FAX1_ASAP7_75t_R	1	0.3	33	18	1703	(-, -)
adder/g6153/SN	-	CON->SN	F	FAX1_ASAP7_75t_R	1	0.3	25	14	1717	(-, -)
adder/g3/Y	-	A->Y	R	INVxp33_ASAP7_75t_R	2	3.0	132	71	1788	(-, -)
adder/g6140/SN	-	B->SN	F	FAX1_ASAP7_75t_R	6	2.1	67	99	1888	(-, -)
adder/g5827_2346/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	3	1.4	53	44	1932	(-, -)
adder/g5825/Y	-	A->Y	F	INVx1_ASAP7_75t_R	4	1.5	28	27	1959	(-, -)
adder/g5808_1617/Y	-	A->Y	R	NAND2xp33_ASAP7_75t_R	1	0.3	19	18	1977	(-, -)
adder/g5806_6783/Y	-	B->Y	F	NAND2xp33_ASAP7_75t_R	2	0.7	21	16	1993	(-, -)
adder/g5786_7482/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	1	0.3	16	15	2008	(-, -)
adder/g5780_5122/Y	-	B->Y	F	NAND2xp33_ASAP7_75t_R	1	0.3	15	10	2018	(-, -)
adder/g5774_5526/Y	-	B->Y	R	NAND2xp33_ASAP7_75t_R	1	0.3	18	13	2032	(-, -)
adder/g5770_5107/Y	-	A->Y	F	NAND2xp33_ASAP7_75t_R	1	0.3	12	10	2042	(-, -)
g14/Y	-	A->Y	R	INVxp33_ASAP7_75t_R	1	0.6	31	21	2063	(-, -)
rem_qd_d_reg[3]/D	-	-	R	DFFASRHQNX1_ASAP7_75t_R	1	-	-	0	2063	(-, -)

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
mod_flip_flop		483	60.478	0.000	60.478	<none> (D)
adder	mod15_n32	346	31.682	0.000	31.682	<none> (D)

(D) = wireload is default in technology library

Innovus Modulo 15 Adder µe Flip Flops 45nm



```

# Command: report_timing
#####
Path 1: MET Setup Check with Pin rem_qd_d_reg[2]/CK
Endpoint: rem_qd_d_reg[2]/D (v) checked with leading edge of 'clk'
Beginpoint: b_qd_d_reg[0]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: worst_case
Other End Arrival Time      0.000
- Setup                     0.069
+ Phase Shift               13.000
= Required Time             12.931
- Arrival Time              9.556
= Slack Time                3.375
Clock Rise Edge             0.000
+ Clock Network Latency (Prop) -0.001
= Beginpoint Arrival Time   -0.001

```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
b_qd_d_reg[0]	CK ^			-0.001	3.375
b_qd_d_reg[0]	CK ^ -> Q ^	DFFRHQX1	0.268	0.268	3.643
adder/g5485_2802	A ^ -> C0 ^	ADDFX1	0.139	0.407	3.783
adder/g5484_1617	CI ^ -> C0 ^	ADDFX1	0.205	0.612	3.987
adder/g5483_3680	CI ^ -> C0 ^	ADDFX1	0.212	0.824	4.199
adder/g5481_5526	CI ^ -> C0 ^	ADDFX1	0.213	1.036	4.412
adder/g5473_6417	CI ^ -> C0 ^	ADDFX1	0.211	1.248	4.623
adder/g5472_7410	CI ^ -> C0 ^	ADDFX1	0.209	1.456	4.832
adder/g5471_1666	CI ^ -> C0 ^	ADDFX1	0.211	1.667	5.042
adder/g5469_2883	CI ^ -> C0 ^	ADDFX1	0.210	1.877	5.252
adder/g5460_6131	CI ^ -> C0 ^	ADDFX1	0.210	2.087	5.462
adder/g5458_8246	CI ^ -> C0 ^	ADDFX1	0.211	2.298	5.673
adder/g5456_1705	CI ^ -> C0 ^	ADDFX1	0.210	2.507	5.883
adder/g5453_3680	CI ^ -> C0 ^	ADDFX1	0.208	2.715	6.091
adder/g5444_5477	CI ^ -> C0 ^	ADDFX1	0.208	2.924	6.299
adder/g5443_6417	CI ^ -> C0 ^	ADDFX1	0.209	3.133	6.508
adder/g5442_7410	CI ^ -> C0 ^	ADDFX1	0.209	3.341	6.717
adder/g5440_2346	CI ^ -> C0 ^	ADDFX1	0.214	3.556	6.931
adder/g5431_1881	CI ^ -> C0 ^	ADDFX1	0.212	3.767	7.143
adder/g5429_7098	CI ^ -> C0 ^	ADDFX1	0.209	3.976	7.352
adder/g5426_1705	CI ^ -> C0 ^	ADDFX1	0.209	4.186	7.561
adder/g5423_3680	CI ^ -> C0 ^	ADDFX1	0.209	4.394	7.770
adder/g5423_3680	CI ^ -> C0 ^	ADDFX1	0.209	4.394	7.770
adder/g5413_6417	CI ^ -> C0 ^	ADDFX1	0.210	4.604	7.979
adder/g5411_1666	CI ^ -> C0 ^	ADDFX1	0.209	4.813	8.189
adder/g5409_2883	CI ^ -> C0 ^	ADDFX1	0.210	5.023	8.399
adder/g5406_6161	CI ^ -> C0 ^	ADDFX1	0.211	5.235	8.610
adder/g5397_5122	CI ^ -> C0 ^	ADDFX1	0.211	5.446	8.821
adder/g5395_2802	CI ^ -> C0 ^	ADDFX1	0.209	5.655	9.031
adder/g5393_3680	CI ^ -> C0 ^	ADDFX1	0.211	5.867	9.242
adder/g5389_8428	CI ^ -> C0 ^	ADDFX1	0.213	6.079	9.454
adder/g5380_1666	CI ^ -> C0 ^	ADDFX1	0.212	6.291	9.666
adder/g5379_2346	CI ^ -> C0 ^	ADDFX1	0.211	6.502	9.877
adder/g5378_2883	CI ^ -> C0 ^	ADDFX1	0.205	6.707	10.082
adder/g5377_9945	B ^ -> Y v	XNOR2X1	0.172	6.878	10.254
adder/g5373_7482	B v -> Y ^	CLKXOR2X1	0.160	7.038	10.414
adder/g5370_6131	B ^ -> C0 ^	ADDFX1	0.139	7.177	10.553
adder/g5367_5122	B ^ -> C0 ^	ADDFX1	0.223	7.401	10.776
adder/g5365_2802	CI ^ -> S v	ADDFX1	0.312	7.712	11.088
adder/g5362_6783	B v -> S ^	ADDFX1	0.331	8.044	11.419
adder/g5358_6260	B ^ -> C0 ^	ADDFX1	0.229	8.272	11.648
adder/g5357_5107	CI ^ -> C0 ^	ADDFX1	0.211	8.484	11.859
adder/g5356_2398	CI ^ -> C0 ^	ADDFX1	0.209	8.693	12.068
adder/g5349_1666	CI ^ -> S ^	ADDFX1	0.248	8.941	12.316
adder/g5346_9945	B ^ -> C0 ^	ADDFX1	0.171	9.111	12.487
adder/g5340_7482	A1 ^ -> Y ^	OA21X1	0.165	9.277	12.652
adder/g5331_2802	B0 ^ -> Y v	OAI31X1	0.092	9.369	12.744
adder/g5328_6783	B v -> Y ^	NAND2X1	0.061	9.430	12.806
adder/g5323_5107	C0 ^ -> Y v	OAI221X1	0.125	9.556	12.931
rem_qd_d_reg[2]	D v	DFFRHQX1	0.000	9.556	12.931

LEC Modulo 15 Adder με Flip Flops

Verification Report	
Category	Count
1. Non-standard modeling options used:	0
2. Incomplete verification:	0
3. User modification to design:	0
4. Conformal Constraint Designer clock domain crossing checks recommended:	0
5. Design ambiguity:	0
6. Compare Results:	PASS

Xcelium Modulo 15 Adder με Flip Flops

